

Assignment 5: ADF Pipeline 2

[Azurelib Academy](#)

1. Create a pipeline name 'Ingestion_Customer_SQLDB_ADLS' which will copy all the records from Customer table to ADLS account as CSV File.
2. Create a pipeline name 'Ingestion_Customer_SQLDB_ADLS_Folder' which will copy all the records from Address table to ADLS account inside the 'Address' folder as CSV File.
3. Create a Pipeline name 'Ingestion_Customer_JOIN_ADLS' which will copy the all the customer names along with their address into the csv. (Hint Join Customer+ Customer Address table)
4. Create a pipeline name 'Ingestion_Product_To_JSON' which will copy all the product records as JSON only if total number of records >10.

Tough Question

5. Create a pipeline name 'Ingestion_Product_Addres_To_JSON' which will copy all the product records as JSON only if total number of records >10. After that check total record count in address table if they are greater than 100 then copy the adress table data as CSV in ADLS.

.....

Assignment 6: ADF Pipeline 3

[Azurelib Academy](#)

Easy Questions:

1. Create a pipeline name 'Without_Foreach_Example' to copy the Customer & Customer address table data into ADLS without using foreach activity.
2. Create a pipeline name 'Foreach_Example' to copy the Customer & Customer address table data into ADLS using the one copy activity.

Tricky Questions:

3. Create a pipeline name 'Foreach_Example_2' which solves the following business use case.

Customer data is very important for our business. Hence whenever we have more than 100 records in the customer table,

we copy the customer data to another table customer_copy within the sql db.

However whenever we do this copy, we first truncate the table 'customer_copy' and then copy the data from 'customer' table

Hint : Use ADF to solve this problem.

4. Business wants to move the data from 3 different tables (Customer, Product, CustomerAddress) to ADLS location in CSV format.

However they want to make a pipeline in such a manner that All these copy happens through an individual pipeline, which is called by

one common pipeline.

5. In the above case problem is that every time the pipeline runs it will overwrite the data. Ensure that all the data goes into

Folder like (Customer/Year/Month/Day) to avoid any overwriting. Please rewrite the pipeline.

6. I am attaching a file, load this file into your ADLS location. This file contains a threshold value. Create a pipeline

in such a manner that, it will first get the threshold value from this file and then check if the record count in the customer

table is more than it or not. If yes then copy the customer data from SQL db to ADLS location in JSON format.

Assignment 5:

1. Create a pipeline name 'Ingestion_Customer_SQLDB_ADLS' which will copy all the records from Customer table to ADLS account as CSV File.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' expanded, listing several pipelines including 'Ingestion_Customer_SQLDB_ADLS'. The main workspace shows the 'Activities' pane with a search bar and a list of activity types: 'Move and transform', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', and 'Data Lake Analytics'. The 'Copy data' activity is selected, and its configuration pane is visible on the right. The configuration includes a 'Name' field set to 'Copy Customer table to ADLS CSV', a 'Description' field with the text 'Copy all the records from Customer table to ADLS account as CSV File.', and an 'Activity state' set to 'Activated'. Other fields like 'Timeout', 'Retry', and 'Retry interval' are also visible.

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 6
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
- Change Data Capture (preview) 0
- Datasets 6
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_Product_Table_Dataset
- Data flows 0
- Power Query 0

SQL Customer Table Dataset

Azure SQL Database
SQL_Customer_Table_Dataset

Connection Schema Parameters

Linked service * SQL_DB_Linked_service1 Test connection Edit + New Learn more

Table SalesLT.Customer Refresh Preview data

☐ Enter manually

https://adf.azure.com/en/authoring/dataset/ADLS_Customer_CSV_Dataset?factory=%2Fsubscriptions%2Fb6db331f-8c4b-42ce-905a-9456c96a03c9%2FresourceGroups%2FAzurelib-Academ

Microsoft Azure Data Factory mission100Gopi01012024adf Search factory and documentation

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 6
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
- Change Data Capture (preview) 0
- Datasets 6
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_Product_Table_Dataset
- Data flows 0
- Power Query 0

ADLS Customer CSV Dataset

DelimitedText
ADLS_Customer_CSV_Dataset

Connection Schema Parameters

Linked service * ADLS_Azure_DataLakeStorage1_Linked_service1 Test connection Edit + New Learn more

File path * landing / CSV / SalesLT.Customer.txt Browse

Compression type Select...

Column delimiter Comma (,)

Row delimiter Default (\r\n, or \n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

First row as header ☒

Null value

SQL_Customer_Table... • ADLS_Customer_CSV... ADLS_Datalake_stora...

Validate Validate copy runtime Debug Add trigger

Copy data

Copy Customer table to ADLS CSV

General Source Sink Mapping Settings User properties

Source dataset *

SQL_Customer_Table_Dataset

Open + New Preview data Li

Use query

Table Query Stored procedure

Query *

select * from SalesLT.Customer

Edit

Query timeout (minutes) ⓘ

120

Isolation level ⓘ

Select...

Partition option ⓘ

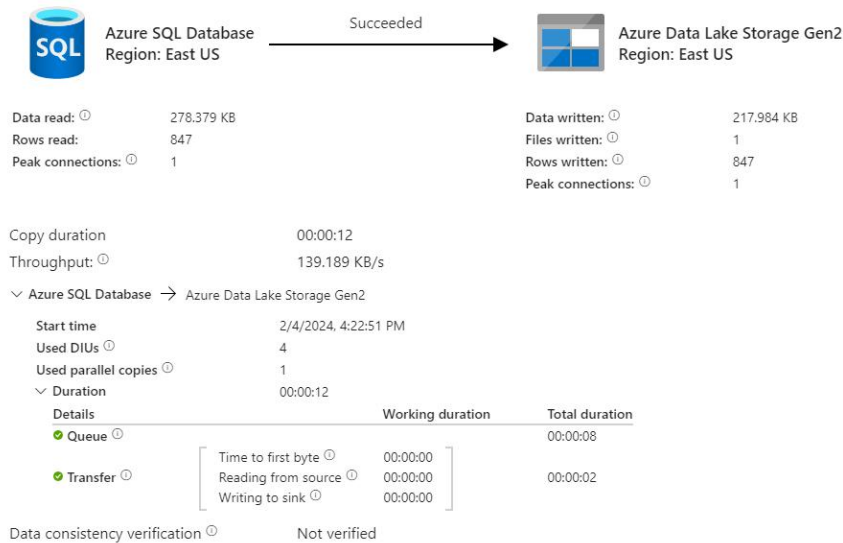
None Physical partitions of table ⓘ Dynamic range ⓘ

Please preview data to validate the partition settings.

Details Refresh

[Learn more on copy performance details from here.](#)

Activity run id: e5ca751a-8430-42f3-801c-7f9e45264881



Published

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Preview experience: Off

Factory validation output: Failed to publish change(s) because of the following validation error(s). Fix error(s) to continue publishing.

- Ingestion_Customer_SQLDB_ADLS_Folder: Pipeline 'Ingestion_Customer_SQLDB_ADLS_Folder' should have at least one Activity.

Factory Resources:

- Pipelines: 6
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS_Folder
- Change Data Capture (preview): 0
- Datasets: 7
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_DataLake_storage_CSV_Dataset
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows: 0
- Power Query: 0

Activities:

- Move and transform
 - Copy data
- Azure Data Explorer
 - Azure Data Explorer C...
- Azure Function
 - Azure Function
- Batch Service
 - Custom
- Databricks
 - Notebook
- IoT
 - IoT
- Python
 - Python
- Data Lake Analytics
 - U-SQL
- General
 - Execute Pipeline

Copy data activity details:

Parameters: Variables: Settings: Output:

Pipeline run ID: 6692db7e-354e-4a32-b121-1403606ccc7

Pipeline status: Succeeded

All status: Monitor in Azure Metrics | Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integra
Copy Custom...	Succeeded	Copy data	2/4/2024, 4:22:50 PM	15s	AutoRe

2. Create a pipeline name 'Ingestion_Customer_SQLDB_ADLS_Folder' which will copy all the records from Address table to ADLS account inside the 'Address' folder as CSV File.

Solution:

We are using Parameterization for creating Dynamic Table , For accessing Source schema and Table and also for pulling into Destination folder

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines7

Copy_Customer_JSON_To_Folder

Copy_Customer_Table_To_CSV

Copy_Customer_To_CSV_Pipe

Copy_Customer_To_JSON

Copy_ProductTable_To_CSV

Ingestion_Customer_SQLDB_ADLS

Ingestion_Customer_SQLDB_ADLS_F...

Change Data Capture (preview)0

Datasets8

ADLS_Customer_CSV_Dataset

ADLS_Customer_CSV_Pipe

ADLS_Datalake_storage_CSV_Dataset

ADLS_Generic_CSV_DS

ADLS_JSON_Dataset

SQL_Customer_Table_Dataset

SQL_DB_Generic_DS

SQL_Product_Table_Dataset

Data flows0

SQL_Customer_Table...

SQL_DB_Generic_DS

Ingestion_Custome...

ADLS_Generic_CSV_DS

Azure SQL Database

SQL_DB_Generic_DS

Connection

Schema

Parameters

New

Delete

☐	Name	Type	Default value	
☐	schemaName	String	Value	
☐	tableName	String	Value	

SQL_Customer_Table...

SQL_DB_Generic_DS

Ingestion_Custome...

ADLS_Generic_CSV_DS

Azure SQL Database

SQL_DB_Generic_DS

Connection

Schema

Parameters

Linked service *

SQL_DB_Linked_service1

Test connection

Edit

New

Learn more

Table

@dataset().schemaName

@dataset().tableName

Preview data

☒ Enter manually

SQL_Customer_Table...

SQL_DB_Generic_DS

Ingestion_Custom...

ADLS_Generic_CSV_DS

7

SQL

Azure SQL Database

SQL_DB_Generic_DS

0

8

set

Connection

Schema

Parameters

Linked service *

SQL_DB_Linked_service1

Test connection

Connection successful

Table

@dataset().schemaName

@dataset().tableName

Enter manually

SQL_DB_Generic_DS

Ingestion_Custom...

ADLS_Generic_CSV_DS

✓ Validate

✓ Validate copy runtime

▶ Debug

⚡ Add trigger

Copy data

Copy records

Address table to...

General

Source

Sink

Mapping

Settings

User properties

Name *

Copy records Address table to ADLS acc...

Learn more

Description

copy all the records
from Address table to ADLS account inside
the 'Address' folder as CSV File.

Activity state

Activated

Deactivated

Timeout

0.12:00:00

Retry

0

Retry interval (sec)

30

Secure output

SQL_DB_Generic_DS • Ingestion_Custom... • ADLS_Generic_CSV_DS •

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Copy data ✓

Copy records
Address table to...

✓
✗
→

General Source Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS Open + New Preview data Learn more

Dataset properties ⓘ

Name	Value
schemaName	SalesLT
tableName	Address

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) ⓘ 120

Isolation level ⓘ Select...

Partition option ⓘ ☒ None ☐ Physical partitions of table ⓘ ☐ Dynamic range ⓘ

SQL_Customer_Table...

SQL_DB_Generic_DS

Ingestion_Custom...

ADLS_Generic_CSV...

DelimitedText

ADLS_Generic_CSV_DS

Connection

Schema

Parameters

+ New

Delete

Name

folderName_param

Type

String

Default value

Value

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory

Validate all

Publish all

Preview experience

Factory Resources

Filter resources by name

Pipelines7

ADLS_Customer_JSON_To_Folder

ADLS_Customer_Table_To_CSV

ADLS_Customer_To_CSV_Pipe

ADLS_Customer_To_JSON

ADLS_ProductTable_To_CSV

Ingestion_Customer_SQLDB_ADLS

Ingestion_Customer_SQLDB_ADLS_F...

Change Data Capture (preview)0

Datasets8

ADLS_Customer_CSV_Dataset

ADLS_Customer_CSV_Pipe

ADLS_Datalake_storage_CSV_Dataset

ADLS_Generic_CSV_DS

ADLS_JSON_Dataset

SQL_Customer_Table_Dataset

SQL_DB_Generic_DS

SQL_Product_Table_Dataset

Data flows0

Power Query0

DelimitedText

ADLS_Generic_CSV_DS

Connection

Schema

Parameters

Linked service

ADLS_Azure_DatalakeStorage1_Link...

Test connection

Connection successful

File path

landing

@dataset().folderName_param

File name

Browse

Preview data

Compression type

Select...

Column delimiter

Comma (,)

Row delimiter

Default (r/n, or v/vn)

Encoding

Default(UTF-8)

Quote character

Double quote (")

Escape character

Backslash (\)

First row as header

☒

Null value

Properties

General

Related

Name

ADLS_Generic_CSV_DS

Description

Annotations

New

Microsoft Azure Data Factory | mission100Gop01012024adff

Factory Resources

- Pipelines
- Datasets
- Data flows
- Power Query

ADLS_Generic_CSV_DS

DelimitedText
ADLS_Generic_CSV_DS

Connection Schema Parameters

Linked service: ADLS_Azure_DataLakeStorageLink... Test connection Connection successful

File path: landing @dataset().folderName_param

Compression type: Select...

Column delimiter: Comma (,)

Row delimiter: Default (\n, or \r\n)

Encoding: Default(UTF-8)

Quote character: Double quote (")

Escape character: Backslash (\)

First row as header: ☒

Null value:

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and parameters.

@dataset().folderName_param

folderName_param

Preview data

Linked service: SQL_DB_Linked_service1

Object: @dataset().schemaName.@dataset().tableName

	AddressID	AddressLine1	AddressLine2	City	StateProvince	CountryRegion	PostalCode	rowguid
1	9	8713 Yosemite Ct.		Bothell	Washington	United States	98011	268af76d7-9441-144fd
2	11	1318 Lasalle Street		Bothell	Washington	United States	98011	981b3aca2-9a96-fb670
3	25	9178 Jumping St.		Dallas	Texas	United States	75201	c8df348f0-a8dd-14a67
4	28	9228 Via Del Sol		Phoenix	Arizona	United States	85004	12ae5fc3e-9b92-3b970
		76010 Indale						84a952222-

Use query

☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) ⓘ

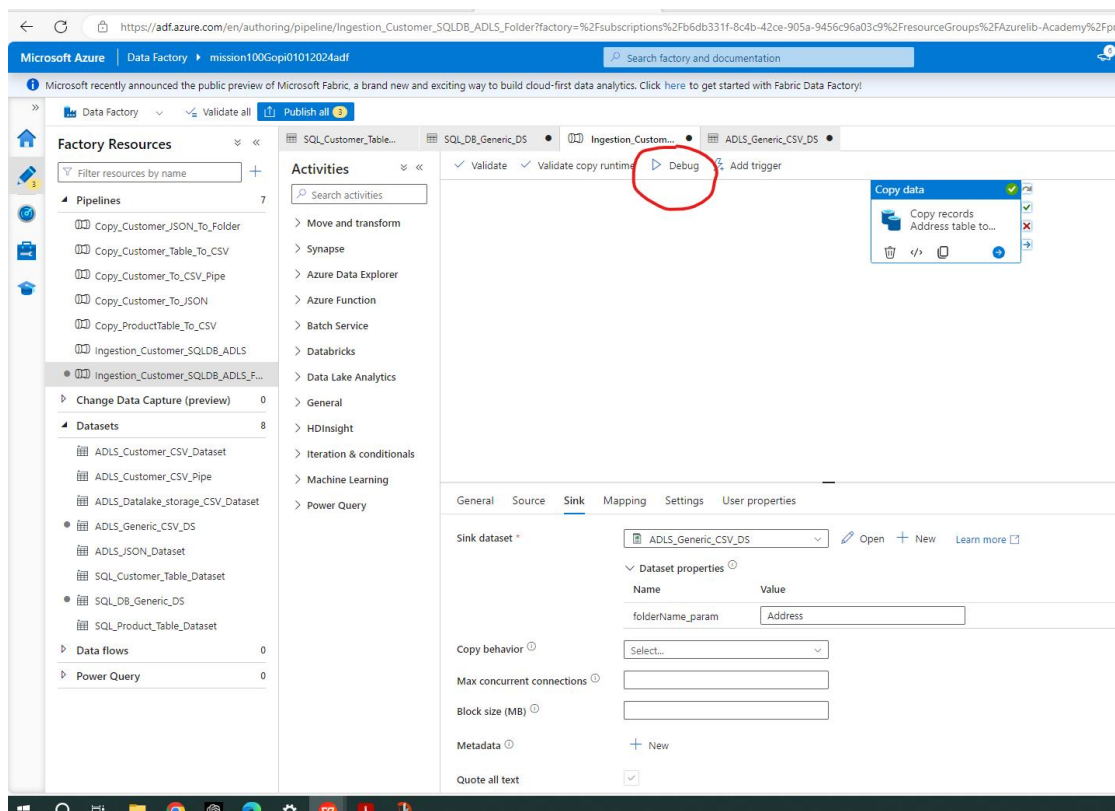
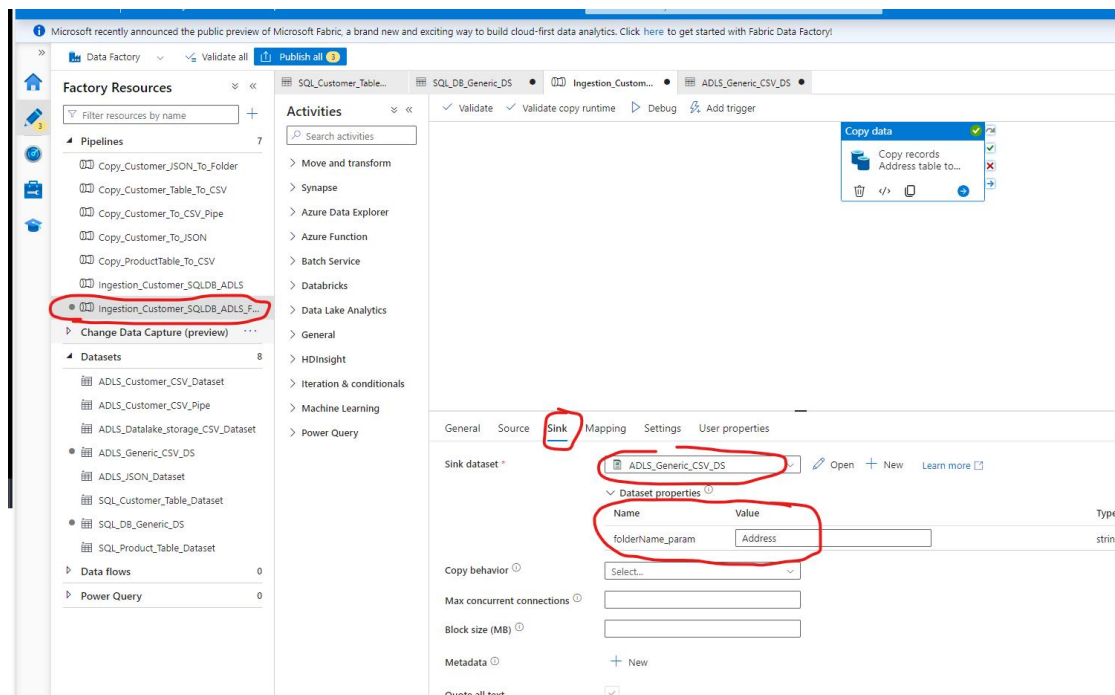
120

Isolation level ⓘ

Select...

Partition option ⓘ

☒ None ☐ Physical partitions of table ⓘ ☐ Dynamic range ⓘ



Microsoft Azure | Data Factory | mission100Gop01012024adf | Search factory and documentation

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

- Pipelines (7)
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_F...
- Change Data Capture (preview) (0)
- Datasets (8)
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_DataLake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows (0)
- Power Query (0)

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Copy data

Copy records Address table to...

Parameters Variables Settings **Output**

Pipeline run ID: e3794cfb-0b9e-447c-9a65-69cfd4df2560 Pipeline status: Succeeded View debug run consumpt

All status

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Copy records Address table t...	Succeeded	Copy data	2/4/2024, 6:50:00 PM	14s	AutoResolveIntegration

Details

Refresh

Learn more on copy performance details from here.

Activity run id: 5649f7d4-8b59-4d04-b2b0-4c02b72d2386

SQL

Azure SQL Database
Region: East US

Succeeded

Azure Data Lake Storage Gen2
Region: East US

Data read: ⓘ

60.192 KB

Rows read:

450

Peak connections: ⓘ

1

Data written: ⓘ

62.727 KB

Files written: ⓘ

1

Rows written: ⓘ

450

Peak connections: ⓘ

1

Copy duration

00:00:12

Throughput: ⓘ

30.096 KB/s

▼ Azure SQL Database → Azure Data Lake Storage Gen2

Start time

2/4/2024, 6:50:00 PM

Used DIUs ⓘ

4

Used parallel copies ⓘ

1

▼ Duration

00:00:12

Details

Working duration

Total duration

● Queue ⓘ

00:00:07

● Transfer ⓘ

Time to first byte ⓘ

00:00:00

Reading from source ⓘ

00:00:00

Writing to sink ⓘ

00:00:00

00:00:02

Data consistency verification ⓘ

Not verified

Home > Storage accounts > gopi01012024ads | Containers >

landing

Container

Search

Upload

Add Directory

Refresh

Rename

Delete

Change tier

Acquire lease

Break lease

Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / Address

Search blobs by prefix (case-sensitive)

Show deleted objects

Name

Modified

Access tier

Archive status

Blob type

Size

Lease state

[-]

[-]

SalesLTAddress.txt

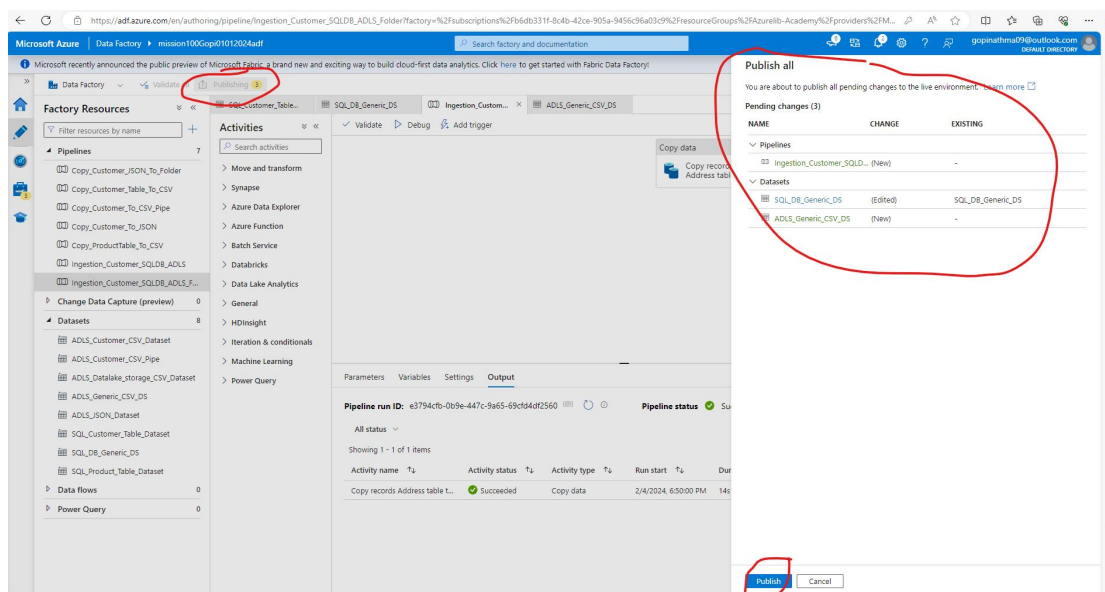
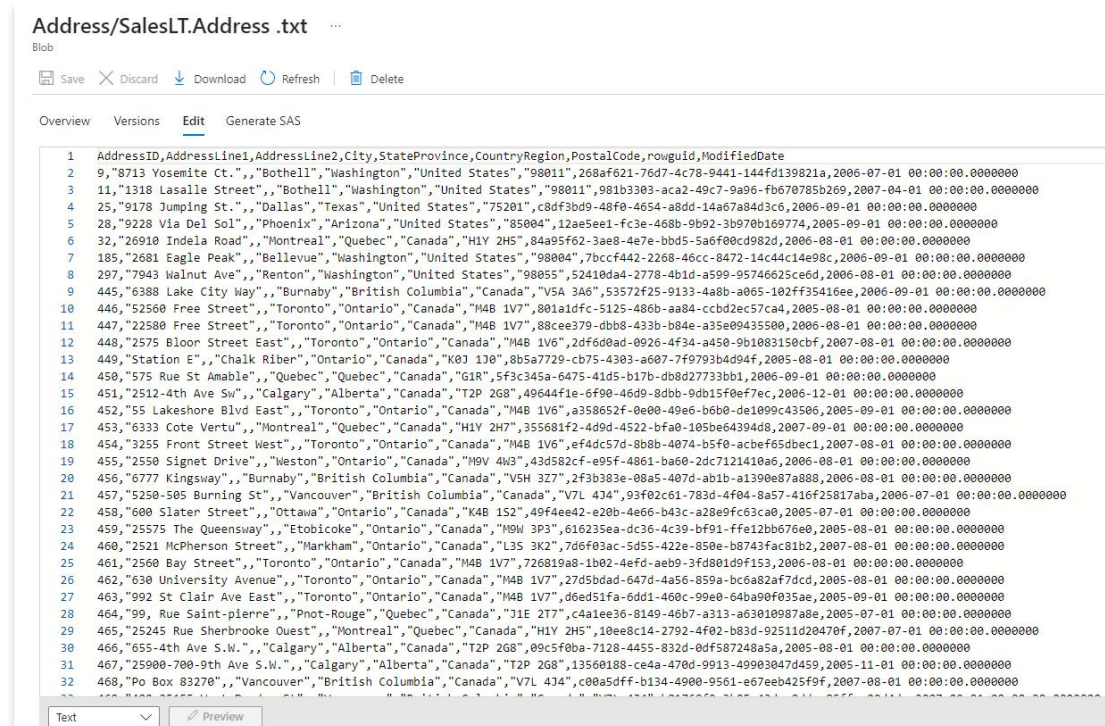
2/4/2024, 6:50:12 PM

Hot (Inferred)

Block blob

61.26 KiB

Available



3. Create a Pipeline name 'Ingestion_Customer_JOIN_ADLS' which will copy the all the customer names along with their address into the csv. (Hint Join Customer+ Customer Address table)

.....

4. Create a pipeline name 'Ingestion_Product_To_JSON' which will copy all the product records as JSON only if total number of records >10.

Approach:

```
IF (TotalProductRecordCount > 10) {  
    Copy Product to JSON  
}
```

Step1: For "TotalProductRecordCount" --> Use "Lookup" Activity to fetch

Step 2: For Comparison --> use "IF" Activity Condition and "Greater" Function in ADF

Step3: For Copy - use "Copy" Activity with Source as SQL Product DS, and Sink as ADLS JSON Dataset

The screenshot shows the 'Properties' pane in the Azure Data Factory (ADF) interface. The 'General' tab is selected. The 'Name' field is set to 'Assignment_5_4_Ingestion_Product_To_JSON'. The 'Description' field contains the text: 'copy all the product records as JSON only if total number of records > 10.' The 'Annotations' section shows a '+ New' button. On the left side of the interface, there is a vertical toolbar with icons for search, add, zoom, and other navigation functions.

Lookup

.....

mission100Gopi01012024adf

Search factory and documentation

he public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Validate all

Publish all

Ingestion_Product_...

ADLS_JSON_Dataset

Assignment_9_4_In...

SQL_Product_Table_...

ADLS_Generic_CSV_DS

ADLS_Generic_JSON

Activities

look

General

Lookup

Lookup

Get Product record count

If Condition

If Check record count gt 10

True

Copy data Product...

False

General

Settings

User properties

Name *

Get Product record count

Learn more

Description

Lookup total number of Product Table records

Activity state

Activated

Deactivated

Timeout

0.12:00:00

Retry

0

Retry interval (sec)

30

Secure output

Secure input

.....

ValidateDebugAdd trigger

Lookup

Get Product record count

Copy data Product...

+

If Condition

If Check record count gt 10

True

Copy data Product...

+

False

+

GeneralSettingsUser properties

Source dataset *SQL_Product_Table_Dataset

First row only

Use queryTableQueryStored procedure

Query *SELECT COUNT(*) AS TotalRecordCount FROM SalesLT.Product

Query timeout (minutes) 120

Isolation level Select...

Partition option NonePhysical partitions of tableDynamic range

Please preview data to validate the partition settings.



Activities

look

General

Lookup

Lookup

Get Product record count

If Condition

If Check record count gt 10

True

Copy data Product...

False

General Settings User properties

Source dataset * SQL_Product_Table_Dataset

First row only

Use query

Query * SELECT COUNT(*) AS TotalRecordCount FROM SalesLT.Product

Query timeout (minutes) 120

Isolation level Select...

Partition option None

Please preview data to validate the partition settings.

Validate Debug Add trigger

Preview data

Linked service: SQL_DB_Linked_service1

Object:

	TotalRecordCount
1	295

.....

Lookup

 Get Product record count

✓

✓

Output

Copy to clipboard

1

2

3

{
 "firstRow": {
 "TotalRecordCount": 295
 },
 "effectiveIntegrationRuntime":
 "AutoResolveIntegrationRuntime (East US)",
 "billingReference": {
 "activityType": "PipelineActivity",
 "billableDuration": [

↗

×

↺

↻

ⓘ

type ↑↓ Run :

.....

ublish all

Ingestion_Product_... ADLS_JSON_Dataset Assignment_5_4_In... SQL_Product_Table_... ADLS_Generic_CSV_DS ADLS_Generic_JSON_...

Activities

IF

Iteration & conditionals

If Condition

Lookup

Get Product record count

If Condition

If Check record count gt 10

True

Copy data Product...

False

General

Activities (1)

User properties

Name *

If Check record count gt 10

Description

If Check record count greater than 10

Activity state

Activated

Deactivated

mission100Gopi0101 X | landing - Microsoft X | mission100Gopi0101 X | if condition sql when X | SQL IF Statement in X | sql - Using IF ELSE st X | sql server - count with X | ChatGPT

s://adf.azure.com/en/authoring/pipeline/Assignment_5_4_Ingestion_Product_To_JSON?factory=%2Fsubscriptions%2F6db331f-8c4b-42ce-905a-9456c96a03c9%2FresourceGroups%2FAzureLib-Academy%2Fproviders

ta Factory mission100Gopi01012024adf

Search factory and documentation

announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Validate all Publish all

Ingestion_Product_... ADLS_JSON_Dataset Assignment_5_4_In... SQL_Product_Table_... ADLS_Generic_CSV_DS ADLS_Generic_JSON_...

Activities

IF

Iteration & conditionals

If Condition

Lookup

Get Product record count

If Condition

If Check record count gt 10

True

Copy data Product...

False

General

Activities (1)

User properties

Expression

@greater(activity('Get Product recor...

Case

Activity

True

Copy data Produ...

1 Activity

False

No activities

ay to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

ADLS_JSON_Dataset

Assignment_5_4_In...

SQL_Product_Table...

ADLS_Generic_CS

✓ Validate ▶ Debug ⚙ Add trigger

Lookup

Get Product record count

If Condition

If Che count

True

Copy data Product...

False

+

General

Activities (1)

User properties

Expression ⓘ

@greater(activity('Get Product recor...

Case

Activity

True

Copy data Produ...

False

1 Activity

No activities

Pipeline expression builder

✓

Add dynamic content below using any combination of expressions, functions and system variables.

@greater(activity('Get Product record count').output.firstRow.TotalRecordCount, 10)

Clear contents

Activity outputs Parameters System variables Functions Variables

greate

⌵ Collapse all

Logical Functions

and

Returns true if both of the parameters are true. Both arguments need to be Booleans. The followin...

greater

Returns true if the first argument is greater than the second. Note, values can only be of type integ...

greaterOrEquals

Returns true if the first argument is greater than or equal to the second. Note, values can only be o...

or

Returns true if either of the parameters are true. Both arguments need to be Booleans. The followi...

OK Cancel

1

2

3

4

.....

Lookup

 Get Product record count

Output

Copy to clipboard

1

2

3

{

"firstRow": {

"TotalRecordCount": 295

},

"effectiveIntegrationRuntime":

"AutoResolveIntegrationRuntime (East US)",

"billingReference": {

"activityType": "PipelineActivity",

"billableDuration": {

type ↑↓ Run :

Pipeline expression builder



Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@greater(activity('Get Product record count').output.firstRow.  
TotalRecordCount, 10)
```

Handwritten annotations: A red line under 'TotalRecordCount' with an arrow pointing to '4'. An arrow points from '10' to '5'. An arrow points from the closing parenthesis ')' to '3'.

[Clear contents](#)

Activity outputs

Parameters

System variables

Functions

Variables

Search

Copy data Product records to JSON
Copy data Product records to JSON activity output

Copy data Product records to JSON
Copy data Product records to JSON pipeline return value

Get Product record count
Get Product record count activity output

Get Product record count
Get Product record count pipeline return value

→ Get Product record count first row
Data of the first row

2

OK

Cancel

.....

ADLS_JSON_Dataset | Assignment_5_4_In... | SQL_Product_Table... | ADLS_Generic_CSV_DS | ADLS_Generic_JSON...

Validate | Debug | Add trigger

Lookup
Get Product record count

If Condition
If Check record count gt 10

True
Copy data Product...

False

General | **Activities (1)** | User properties

Expression [ⓘ] `@greater(activity('Get Product record count'), 10)`

Case	Activity
True	Copy data Product... 1 Activity
False	No activities

Data Factory | mission100cop10101202450 | Search factory and documentation

Recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory | Validate all | Publish all

My Resources

Resources by name

Items 9

- Copy_Customer_JSON_To_Folder
- Copy_Customer_Table_To_CSV
- Copy_Customer_To_CSV_Pipe
- Copy_Customer_To_JSON
- Copy_ProductTable_To_CSV
- Ingestion_Customer_SQLDB_ADLS
- Ingestion_Customer_SQLDB_ADLS_Folder
- Ingestion_Product_To_JSON
- Assignment_5_4_Ingestion_Product_To_JSON

Activities

copy

Move and transform

Copy data

Assignment_5_4_Ingestion_Product_To_JSON > If Check record count gt 10 > True activities

Copy data Product records to JSON

of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Publish all

Activities

copy

Move and transform

Copy data

Assignment_5_4_Ingestion_Product_To_JSON > If Check record count gt 10 > True activities

Copy data

Copy data Product records to JSON

General

Name * Copy data Product records to JSON

Description

Activity state ☒ Activated ☐ Deactivated

Timeout 0:12:00:00

Retry 0

Retry interval (sec) 30

Secure output ☐

Secure input ☐

Publish all

Activities

copy

Move and transform

Copy data

Assignment_5_4_Ingestion_Product_To_JSON > If Check record count gt 10 > True activities

Copy data

Copy data Product records to JSON

General

Source

Source dataset * SQL_Product_Table_Dataset

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Partition option ☒ None ☐ Physical partitions of table ☐ Dynamic range

Please preview data to validate the partition settings.

Additional columns + New

«

✓ Validate

✓ Validate copy runtime

▶ Debug

⚡ Add trigger

Assignment_5_4_Ingestion_Product_To_JSON > If Check record count gt 10 > True activities

Copy data

Copy data Product records to JSON

✓

✗

↶

↷

General

Sink

Mapping

Settings

User properties

Sink dataset *

ADLS_Generic_JSON_DS

Open

New

Learn more

Dataset properties

Name	Value
folderName_param	Product_JSON

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Methods

New

.....

ADLS_JSON_DatasetAssignment_5.4.In...SQL_Product_Table...ADLS_Generic_CSV_DSADLS_Generic_JSON...

«ValidateValidate copy runtimeDebugAdd trigger

Assignment_5.4.In...Check record count gt 10 > True activities

Trigger test runs of the current pipeline without publishing your changes to the service.

Copy dataCopy data Product records to JSON

GeneralSourceSinkMappingSettingsUser properties

Sink dataset *ADLS_Generic_JSON_DSOpen+NewLearn more

Dataset properties

Name	Value
folderName_param	Product_JSON

Copy behaviorSelect...

Max concurrent connections

Block size (MB)

Metadata+ New

File pattern

Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Publish allPreview experience

«Ingestion_Product...ADLS_JSON_DatasetAssignment_5.4.In...SQL_Product_Table...ADLS_Generic_CSV_DSADLS_Generic_JSON...

ActivitiesValidateDebugAdd trigger

copy

Move and transformCopy data

LookupGet Product record count

If ConditionIf Check record count gt 10

TrueCopy data Product...

False

ParametersVariablesSettingsOutput

Pipeline run ID: 19474e48-38cf-4b8a-8cb5-384d40863ed5Pipeline status SucceededView debug run

All statusListMonitor in Azure MetricsExport

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data Product records to...	Succeeded	Copy data	2/6/2024, 12:24:22 AM	14s	AutoResolveIntegration		01cf8a5-793
If Check record count gt 10	Succeeded	If Condition	2/6/2024, 12:24:21 AM	16s			9ba3bea8-e6
Get Product record count	Succeeded	Lookup	2/6/2024, 12:24:15 AM	5s	AutoResolveIntegration		c22f0b3d-16c

✓ Validate ▶ Debug ⚡ Add trigger

Lookup

Get Product record count

Lookup total number of Porduct Table records

Activity state ^① ☒ Activated ☐ Deactivated

Timeout ^① 0.12:00:00

Retry ^① 0

If Condition

If Check record count gt 10

True

Copy data Product...

False

General Settings User properties

Name * Get Product record count [Learn more](#) ^①

Description Lookup total number of Porduct Table records

.....

```
select * from table
where 10 < (select count(*) from table)
```

-->

```
Select * from SalesLT.Product
WHERE (Select COUNT(*) from SalesLT.Product ) > 10
```

Wrong:

```
IF EXISTS (SELECT * FROM TABLE WHERE COLUMN = 'SOME VALUE')
    --INSERT SOMETHING
ELSE
    --INSERT SOMETHING ELSE
```

Query is:

IF EXISTS (Select COUNT(*) from SalesLT.Product) > 1000)

Select * from SalesLT.Product

OR

```
SELECT * FROM Customers C
WHERE EXISTS
( SELECT 1 FROM
Orders O
WHERE O.CustomerID = C.CustomerID );
```

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. [Click here](#) to get started with Fabric Data Factory!

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines (8)
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
- Change Data Capture (preview) (0)
- Datasets (8)
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows (0)
- Power Query (0)

Activities

cop

Move and transform

Copy data

Copy data

Copy Product records as JSON

General | Source | Sink | Mapping | Settings | User properties

Name * Copu Product records as JSON [Learn more](#)

Description copy all the product records as JSON only if total number of records > 10.

Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0

Retry interval (sec) 30

Secure output ☐

Secure input ☐

Microsoft Azure | Data Factory | mission100Gop1012024adf

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Factory Resources

- Pipelines
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
- Change Data Capture (preview) 0
- Datasets 8
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows 0
- Power Query 0

Activities

Move and transform

Copy data

Copy data

Copy Product records as JSON

General **Source** Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS

Dataset properties

Name	Value
schemaName	SalesLT
tableName	Product

Use query ☒ Table ☒ Query ☐ Stored procedure

Query *
Select * from SalesLT.Product
WHERE (Select COUNT(*) from SalesLT.Product) > 10

Query timeout (minutes) 120

Microsoft Azure | Data Factory | mission100Gop1012024adf

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Factory Resources

- Pipelines
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
- Change Data Capture (preview) 0
- Datasets 8
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows 0
- Power Query 0

ADLS_JSON_Dataset

JSON

Connection Schema Parameters

Linked service * ADLS_Azure_DataLakeStorage1_Link

File path * landing / JSON / File name

Compression type None

Encoding Default(UTF-8)

Copy data

Copy Product records as JSON

General **Source** **Sink** Mapping Settings User properties

Sink dataset * ADLS_JSON_Dataset

Copy behavior

Max concurrent connections

Block size (MB)

Metadata

File pattern

Microsoft Azure | Data Factory | mission100Gopi01012024adf

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Factory Resources

- Pipelines (8)
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
- Change Data Capture (preview) (0)
- Datasets (8)
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_JSON_Dataset
 - SQL_Customer_Table_Dataset
 - SQL_DB_Generic_DS
 - SQL_Product_Table_Dataset
- Data flows (0)
- Power Query (0)

Activities

- Copy data

Copy data

Copy Product records as JSON

Sink dataset * ADLS_JSON_Dataset

Copy behavior Select...

Max concurrent connections

Block size (MB)

Metadata + New

File pattern

Copy data

Copy Product records as JSON

Parameters Variables Settings **Output**

Pipeline run ID: a4f82184-2754-433d-9600-e1479cb74b41 Pipeline status **Succeeded** View debug run consumption

All status

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity
Copy Product...	Succeeded	Copy data	2/4/2024, 7:50:16 PM	12s	AutoResolveIntegration		620647

Input

 Copy to clipboard

```
{
  "source": {
    "type": "AzureSqlSource",
    "sqlReaderQuery": "Select * from SalesLT.Product \nWHERE (Select COUNT(*) from SalesLT.Product ) > 10",
    "queryTimeout": "02:00:00",
    "partitionOption": "None"
  },
  "sink": {
    "type": "JsonSink",
    "storeSettings": {
      "type": "AzureBlobFSWriteSettings"
    },
    "formatSettings": {
      "type": "JsonWriteSettings"
    }
  },
  "enableStaging": false
}
```

Output

 Copy to clipboard [Learn more on output details](#) 

```
{
  "dataRead": 715233,
  "dataWritten": 998563,
  "filesWritten": 1,
  "sourcePeakConnections": 1,
  "sinkPeakConnections": 1,
  "rowsRead": 295,
  "rowsCopied": 295,
  "copyDuration": 10,
  "throughput": 357.616,
  "errors": [],
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (East US)",
  "usedDataIntegrationUnits": 4,
  "billingReference": {
    "activityType": "DataMovement",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.06666666666666667,
        "unit": "DIUHours"
      }
    ]
  },
  "totalBillableDuration": [
    {
      "meterType": "AzureIR",
      "duration": 0.06666666666666667,
      "unit": "DIUHours"
    }
  ]
},
"usedParallelCopies": 1,
"executionDetails": [
  {
    "source": {
      "type": "AzureSqlDatabase",
      "region": "East US"
    },
    "sink": {
      "type": "AzureBlobFS",
      "region": "East US"
    },
    "status": "Succeeded",
    "start": "2/4/2024, 7:50:17 PM",
    "duration": 10,
    "usedDataIntegrationUnits": 4,
    "usedParallelCopies": 1,
  }
]
```

Details [Refresh](#)

[Learn more on copy performance details from here.](#)

Activity run id: 6206471e-b11e-4f77-84bd-9308a90a3fdf

**Azure SQL Database**
Region: East US

Succeeded

**Azure Data Lake Storage Gen2**
Region: East US

Data read: 715.233 KB

Rows read: 295

Peak connections: 1

Data written: 998.563 KB

Files written: 1

Rows written: 295

Peak connections: 1

Copy duration 00:00:10
Throughput: 357.616 KB/s

Azure SQL Database → **Azure Data Lake Storage Gen2**

Start time 2/4/2024, 7:50:17 PM

Used DIUs 4

Used parallel copies 1

Duration 00:00:10

Details	Working duration	Total duration
Queue		00:00:06
Transfer	Time to first byte 00:00:00 Reading from source 00:00:00 Writing to sink 00:00:00	00:00:02

Data consistency verification Not verified

landing - Microsoft | mission100Gop101 | mission100Gop101 | sql query to get all | execute a query in a | List tables in Azure | sql query to check | sql - How to get ro

https://portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBlade/~/overview/storageAccountId/%2Fsubscriptions%2Fb6db331f-8c4b-42ce-905a-9456c96a03c9%2FresourceGroups%2FAzurelib-Academy%2FstorageAccounts%2Fgopi01012024adls/containers/landing

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Storage accounts > gopi01012024adls | Containers >

landing Container

Search Upload Add Directory Refresh Rename Delete Change tier Acquire lease Break lease Give feedback

Overview Authentication method: Access key (Switch to Microsoft Entra user account) Location: landing / JSON Search blobs by prefix (case-sensitive) Show c

Diagnose and solve problems Access Control (IAM)

Settings

Shared access tokens

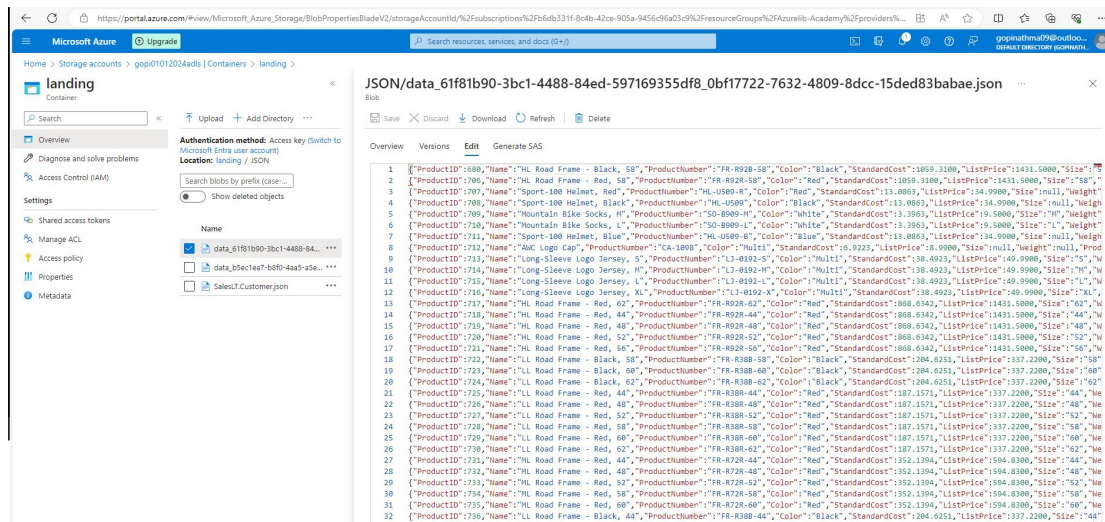
Manage ACL

Access policy

Properties

Metadata

name	Modified	Access tier	Archive status	Blob type
☐ data_61f81b90-3bc1-4488-84e3-997169355df8_0bf17722-7632-4809-8d0c-15ded83...	2/4/2024, 7:42:50 PM	Hot (inferred)		Block blob
☐ data_b5ec1ea7-bbf0-4a55-95e1-1abecb46f623_5f64c65c-2e59-4fcc-8f37-3581fb83c...	2/4/2024, 7:40:48 PM	Hot (inferred)		Block blob
☐ SalesLT.Customer.json	1/28/2024, 10:47:15 PM	Hot (inferred)		Block blob



3. Create a Pipeline name 'Ingestion_Customer_JOIN_ADLS' which will copy the all the customer names along with their address into the csv. (Hint Join Customer+ Customer Address table)

Solution Query:

<https://www.databasestar.com/multiple-inner-joins/>

```
Select C.FirstName, C.MiddleName, C.LastName, A.AddressLine1, A.AddressLine2,
A.City, A.StateProvince, A.CountryRegion, A.PostalCode
FROM SalesLT.Customer C
INNER JOIN SalesLT.CustomerAddress CA ON C.CustomerID = CA.CustomerID
INNER JOIN SalesLT.Address A ON A.AddressID = CA.AddressID
```

The syntax looks like this:

```
SELECT your_columns FROM table1
INNER JOIN table2 ON table1.col1 = table2.col1
INNER JOIN table3 ON table2.col2 = table3.col2;
```


Preview data

Linked service: SQL_DB_Linked_service1

Object: SalesLT.CustomerAddress

	CustomerID	AddressID	AddressType	rowguid	ModifiedDate
1	29485	1086	Main Office	16765338-dbe4-4421-b5e9-3836b9278e63	2007-09-01T00:00:00
2	29486	621	Main Office	22b3e910-14af-4ed5-8b4d-23bbe757414d	2005-09-01T00:00:00
3	29489	1069	Main Office	a095c88b-d7e6-4178-a078-2eca44214801	2005-07-01T00:00:00
4	29490	887	Main Office	f12e1702-d897-4035-b614-0fe2c72168a9	2006-09-01T00:00:00
5	29492	618	Main Office	5b3b3eb2-3f43-47ed-a20c-23697dabf23b	2006-12-01T00:00:00
6	29494	537	Main Office	492d92b6-31af-47ea-8e00-7c11c7ccc20d	2005-09-01T00:00:00
7	29496	1072	Main Office	0a66b0f3-24bc-4148-9661-26e935cec99a	2007-09-01T00:00:00
8	29497	889	Main Office	7e0b56fd-7324-4898-bedb-2f56a843bb0f	2005-07-01T00:00:00
9	29499	527	Main Office	c90cb0c3-976a-4075-a2b9-13136f4f1a92	2006-09-01T00:00:00
10	29502	893	Main Office	8acb1c6a-7cdf-417b-afd5-9f4c642f3c7e	2007-07-01T00:00:00

Connection Schema Parameters

Linked service *

SQL_DB_Linked_service1

Test connection Edit

Connection successful

Table

SalesLT.Address

Refresh Preview data

Enter manually

Preview data

Linked service: SQL_DB_Linked_service1

Object: SalesLT.Address

	AddressID	AddressLine1	AddressLine2	City	StateProvince	CountryRegion	PostalCode	rowguid	ModifiedDate
1	9	8713 Yosemite Ct.		Bothell	Washington	United States	98011	266af621-76d7-4c78-9441-144fd139821a	2006-07-01T00:00:00
2	11	1318 Lasalle Street		Bothell	Washington	United States	98011	981b3303-aca2-49c7-9a96-b670785b269	2007-04-01T00:00:00
3	25	9178 Jumping St.		Dallas	Texas	United States	75201	c9d9b0d9-46f0-4654-a6d0-14a67a94d3c6	2006-09-01T00:00:00
4	28	9228 Via Del Sol		Phoenix	Arizona	United States	85004	12ae5ee1-fc3e-468b-9692-30570b169774	2005-09-01T00:00:00
5	32	26910 Indela Road		Montreal	Quebec	Canada	H1Y 2H5	84a95f62-3a69-4a7e-bb05-5a6f00cd992d	2006-08-01T00:00:00
6	185	2681 Eagle Peak		Bellevue	Washington	United States	98004	7bccc442-2268-46cc-8472-14c44c14e99c	2006-09-01T00:00:00
7	297	7943 Walnut Ave		Renton	Washington	United States	98055	52410d84-2778-4b1d-a599-95746625e6d	2006-08-01T00:00:00
8	445	6388 Lake City Way		Burnaby	British Columbia	Canada	V5A 3A6	53572f25-9133-4a8b-a065-102f935416ee	2006-09-01T00:00:00
9	446	52560 Free Street		Toronto	Ontario	Canada	M4B 1V7	801a1dfc-5125-486b-aa84-cbd26c57ca4	2005-08-01T00:00:00
10	447	22580 Free Street		Toronto	Ontario	Canada	M4B 1V7	88cee379-dbb8-433b-b84e-a35e09435500	2006-08-01T00:00:00

Query:

```
Select C.FirstName, C.MiddleName, C.LastName, A.AddressLine1, A.AddressLine2,
A.City, A.StateProvince, A.CountryRegion, A.PostalCode
FROM SalesLT.Customer C
INNER JOIN SalesLT.CustomerAddress CA ON C.CustomerID = CA.CustomerID
INNER JOIN SalesLT.Address A ON A.AddressID = CA.AddressID
```

The syntax looks like this:

```
SELECT your_columns FROM table1
INNER JOIN table2 ON table1.col1 = table2.col1
INNER JOIN table3 ON table2.col2 = table3.col2;
```

Object: SalesLT.Customer

	CustomerID	NameStyle	Title	FirstName	MiddleName	LastName	Suffix	CompanyName	SalesPerson
1	1	false	Mr.	Orlando	N.	Gee		A Bike Store	adventureworks

SalesLT.Address

AddressID	AddressLine1	AddressLine2	City	StateProvince	CountryRegion
-----------	--------------	--------------	------	---------------	---------------

SalesLT.Customer C

SalesLT.CustomerAddress

	CustomerID	AddressID	AddressType	rowguid	ModifiedDate
1	29485	1086	Main Office	16765338-dbe4-4421-b5e9-3836b9278e63	2007-09-01

Ass_5_3_JOIN_Inge...

«
✓ Validate
✓ Validate copy runtime
▶ Debug
⚡ Add trigger

Copy data

Copy data1
Customer JOIN A...



General

Source

Sink

Mapping

Settings

User properties

Name *

Copy data1 Customer_JOIN_ADLS

[Learn more](#)

Description

Create a Pipeline name
'Ingestion_Customer_JOIN_ADLS' which
will copy the all the customer

Activity state ①

☒ Activated ☐ Deactivated

Timeout ①

0.12:00:00

Retry ①

0

Retry interval (sec) ①

30

Secure output ①

☐

Secure input ①

☐

Ass_5_3_JOIN_Inge...

Validate

Validate copy runtime

Debug

Add trigger

Copy data

Copy data1
Customer_JOIN_A...

General

Source

Sink

Mapping

Settings

User properties

Source dataset *

SQL_DB_Generic_DS

Open

New

Preview

Dataset properties

Name	Value
schemaName	SalesLt
tableName	Customer

Use query

Table

Query

Stored procedure

Query *

FROM SalesLT.Customer C
INNER JOIN SalesLT.CustomerAddress
CA ON C.CustomerID = CA.CustomerID

Edit

Query timeout (minutes)

120

ivities

Validate Validate copy runtime Debug Add trigger

Preview data

Linked service: SQL_DB_Linked_service1

Object:

	FirstName	MiddleName	LastName	AddressLine1	AddressLine2	City	StateProvince	CountryR
1	Jacob	N.	Dean	#9900 2700 Production Way		Burnaby	British Columbia	Canada
2	Lola	M.	McCarthy	1050 Oak Street		Seattle	Washington	United States
3	Peter		Houston	1200 First Ave.		Joliet	Illinois	United States
4	Joseph	P.	Mitzner	123 Camelia Avenue		Oxnard	California	United States
5	Frank		Miller	123 W. Lake Ave.		Peoria	Illinois	United States
6	Donna	F.	Carreras	12345 Sterling Avenue		Irving	Texas	United States
7	John	A.	Beaver	1318 Lasalle Street		Bothell	Washington	United States

Execute Pipeline

Execute SSIS package

Stored procedure

Script

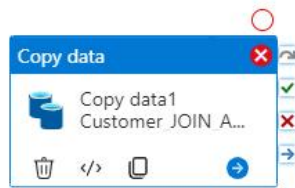
Use query

Query *

FROM SalesLT.Customer C
INNER JOIN SalesLT.CustomerAddress
CA ON C.CustomerID = CA.CustomerID

Edit

Query timeout (minutes) 120



General Source **Sink** Mapping Settings User properties

Sink dataset * ADLS_Generic_CSV_DS Open New [Learn more](#)

Dataset properties ¹

Name	Value
folderName_param	Ass_5_3_Customer_Join_Address_CSV

Copy behavior ¹ Select...

Max concurrent connections ¹

Block size (MB) ¹

Metadata ¹ + New

Ass_5_3_JOIN_Inge... Preview experience Off Refresh

Validate Debug Add trigger Code Help

Copy data ✓

Copy data1 ✓

Customer JOIN A...

Parameters Variables Settings **Output**

Pipeline run ID: 800a5c6e-1c6c-4dba-bb67-fa31e7103c46 Refresh Info **Pipeline status** ✓ Succeeded [View debug run consumption](#)

[Monitor in Azure Metrics](#) [Export to CSV](#)

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data1 Customer_JOIN_...	✓ Succeeded	Copy data	2/10/2024, 2:10:08 AM	12s	AutoResolveIntegration		cb7beb26-3a59-

Input

Copy to clipboard

```
{
  "source": {
    "type": "AzureSqlSource",
    "sqlReaderQuery": "Select C.FirstName, C.MiddleName,
C.LastName, A.AddressLine1, A.AddressLine2, \nA.City,
A.StateProvince, A.CountryRegion, A.PostalCode\nFROM
SalesLT.Customer C\nINNER JOIN SalesLT.CustomerAddress CA ON
C.CustomerID = CA.CustomerID\nINNER JOIN SalesLT.Address A
ON A.AddressID = CA.AddressID",
```

03c46 @ ↺ ⓘ

Activity type ↑↓

Run start

Copy data1 Customer_JOIN_...

✔ Succeeded

Copy data

2/10/2024, 2

Output

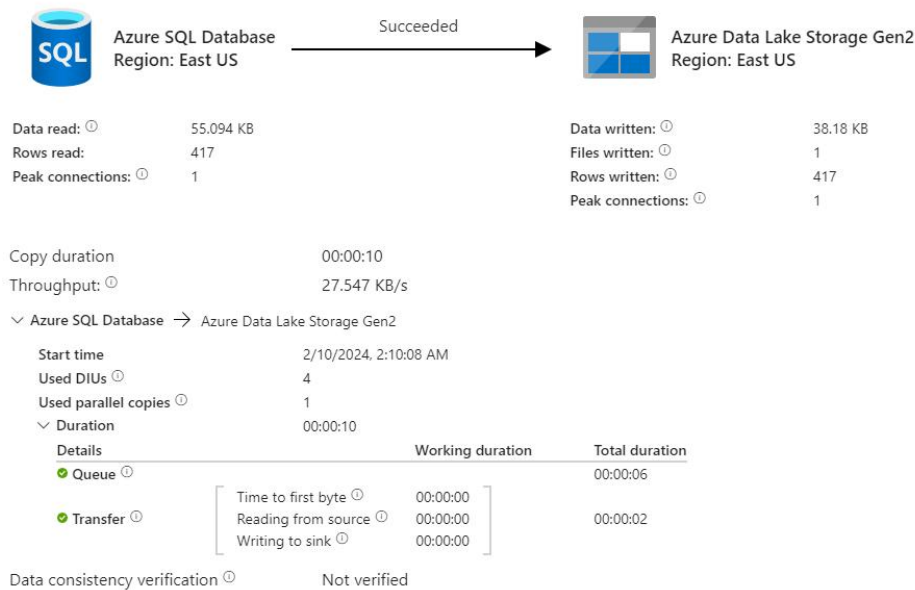
Copy to clipboard [Learn more on output details](#)

```
{
  "dataRead": 55094,
  "dataWritten": 38180,
  "filesWritten": 1,
  "sourcePeakConnections": 1,
  "sinkPeakConnections": 1,
  "rowsRead": 417,
  "rowsCopied": 417,
  "copyDuration": 10,
```

Details Refresh

Learn more on copy performance details from here.

Activity run id: cb7beb26-3a59-41c2-b345-bf9d3e250d84



portal.azure.com/?feature.msalsjs=true#view/Microsoft_Azure_Storage/BlobPropertiesBladeV2/storageAccountId/%2Fsubscriptions%2Fb6db331f-8c4b-42ce-905a-9456c96a03c9%2FresourceGroups%2FAzureLib-... Search resources, services, and docs (G+)

Home > Storage accounts > gopi01012024ads > Containers > landing >

landing Container

Search Upload Add Directory ...

Overview Diagnose and solve problems Access Control (IAM) Settings Shared access tokens Manage ACL Access policy Properties Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: landing / Ass_5_3_Customer_Join_Address_CSV
Search blobs by prefix (case-...)
Show deleted objects

Name
[-] data_cb7beb26-3a59-41c2-b3... ***

Ass_5_3_Customer_Join_Address_CSV/data_cb7beb26-3a59-41c2-b345-bf9d3e250d84_f70
Blob

Save Discard Download Refresh Delete

Overview Versions Edit Generate SAS

```
1 FirstName,MiddleName,LastName,AddressLine1,AddressLine2,City,StateProvince,CountryRegion,PostalCode
2 "Jacob","N.",,"Dean","49900 2700 Production Way",,"Burnaby","British Columbia","Canada","V5A 4X1"
3 "Lola","M.",,"McCarthy",,"1898 Oak Street",,"Seattle","Washington",,"United States",,"98104"
4 "Peter",,"Houston",,"1200 First Ave.",,"Joliet","Illinois",,"United States",,"60433"
5 "Joseph","P.",,"Mitzner",,"123 Camelia Avenue",,"Oxnard","California",,"United States",,"93030"
6 "Frank",,"Miller",,"123 W. Lake Ave.",,"Peoria",,"Illinois",,"United States",,"61600"
7 "Donna",,"F.",,"Carneras",,"12345 Sterling Avenue",,"Irving",,"Texas",,"United States",,"75061"
8 "John",,"A.",,"Beaver",,"1318 Lasalle Street",,"Bothell",,"Washington",,"United States",,"98011"
9 "Peggy",,"J.",,"Justice",,"15 East Main",,"Port Orchard",,"Washington",,"United States",,"98366"
10 "Janet",,"M.",,"Gates",,"165 North Main",,"Austin",,"Texas",,"United States",,"78701"
11 "Barry",,"Johnson",,"2000 300th Street",,"Denver",,"Colorado",,"United States",,"80203"
12 "Suzana De Abreu",,"A.",,"Canuto",,"20225 Lansing Ave",,"Montreal",,"Quebec",,"Canada",,"H1V 2H7"
13 "Stacey",,"M.",,"Cereghino",,"220 Mercy Drive",,"Garland",,"Texas",,"United States",,"75040"
14 "Patricia",,"Doyle",,"225 South 314th Street",,"Federal Way",,"Washington",,"United States",,"98003"
15 "Orlando",,"N.",,"Gee",,"2251 Elliot Avenue",,"Seattle",,"Washington",,"United States",,"98104"
16 "Alan",,"Steiner",,"2255 254th Avenue Se",,"Albany",,"Oregon",,"United States",,"97321"
17 "Michael",,"Brundage",,"22555 Paseo De Las Americas",,"San Diego",,"California",,"United States",,"92102"
18 "Kendra",,"N.",,"Thompson",,"22571 South 2500 East",,"Idaho Falls",,"Idaho",,"United States",,"83402"
19 "Johnny",,"A.",,"Caprio",,"22580 Free Street",,"Toronto",,"Ontario",,"Canada",,"M4B 1V7"
20 "Michelle",,"Alexander",,"22589 West Craig Road",,"North Las Vegas",,"Nevada",,"United States",,"89030"
21 "John",,"Ford",,"23025 S.W. Military Rd.",,"San Antonio",,"Texas",,"United States",,"78204"
22 "John",,"Arthur",,"2345 North Freeway",,"Houston",,"Texas",,"United States",,"77003"
23 "Mary",,"Alexander",,"2345 West Spencer Road",,"Lynnwood",,"Washington",,"United States",,"98036"
24 "Tim",,"Kim",,"25 Danger Street West",,"Florence",,"Toronto",,"Ontario",,"Canada",,"M4B 1V5"
25 "Josh",,"Barnhill",,"25 First Canadian Place",,"Toronto",,"Ontario",,"Canada",,"M4B 1V5"
26 "Ajay",,"Hanchepalli",,"25 Hartfield Road",,"Wimbledon",,"London",,"England",,"United Kingdom",,"SW19 3RU"
27 "Yao-Qiang",,"Cheng",,"25 N State St",,"Chicago",,"Illinois",,"United States",,"60610"
28 "Stephen",,"A.",,"Hsu",,"250 Kingsbridge Garden Circle",,"Mississauga",,"Ontario",,"Canada",,"L5B 3V4"
29 "Vanessa",,"J.",,"Tench",,"2500 N Serene Blvd",,"19th Floor",,"El Segundo",,"California",,"United States",,"90245"
30 "Randall",,"Boseman",,"2500 North Stemmons Freeway",,"Dallas",,"Texas",,"United States",,"75201"
31 "Robin",,"H.",,"McGuigan",,"2500 University Avenue",,"Toronto",,"Ontario",,"Canada",,"M4B 1V5"
32 "Jo",,"Roun",,"250000 Fehr Mile Road",,"Detroit",,"Michigan",,"United States",,"48226"
```

Tough Question

5. Create a pipeline name 'Ingestion_Product_Addres_To_JSON' which will copy all the product records as JSON only if total number of records >10. After that check total record count in address table if they are greater than 100 then copy the address table data as CSV in ADLS

Video : [Azurelib Academy \(freshlearn.com\)](https://www.freshlearn.com/AzurelibAcademy)

Session 06 (Live): Lookup /If / Execute pipeline activity / Dynamic Date

Part 2

Scenario 2:

Load the data from customer table into ADLS location when the total record count in customer table is greater than 500. After this check the record count of product table. If it is greater than 200 then copy the product table as well.

Step 1: To fetch the record count for customer table-- Lookup

Step 2: Compare of record count > 500

Step 3: If yes then copy

Step 4: Fetch the record count for product

Step 5: Compare if count >200

Step 6: If yes then Copy



Scenario 2:

Load the data from customer table into ADLS location when the total record count in customer table is greater than 500. After this check the record count of product table. If it is greater than 200 then copy the product table as well.

```
if ( cust record count > 500) {  
    Copy the data  
    if ( prod record count> 200) {  
        copy the product table  
    }  
}
```



Pipeline 1:

```
if ( cust record count > 500) {  
    Copy the data  
    Call pipeline 2:  
}
```

Pipeline 2: Child pipeline

```
if ( prod record count> 200) {  
    copy the product table  
}
```

Nested If" - is not allowed in ADF: "Solution": is you need to "Break the pipeline" (make it 2 pipeline)
- Instead of 1 Pipeline make it 2 pipeline and make some connection between them

Scenario 2:

Load the data from customer table into ADLS location when the total record count in customer table is greater than 500. After this check the record count product table. If it is greater than 200 then copy the product table as well.

Pipeline 1:

Step 1: To fetch the record count for customer table-- Lookup

Step 2: Compare of record count > 500

Step 3: If yes then copy

Step 4: Call the pipeline 2



Pipeline 2:

Step 4: Fetch the record count for product

Step 5: Compare if count >200

Step 6: If yes then Copy

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory.

Factory Resources

- Assignment_5_4_Ingestion_Product...
- Copy_Customer_JSON_To_Folder
- Copy_Customer_Table_To_CSV
- Copy_Customer_Table_To_CSV_Pipe
- Copy_Customer_To_JSON
- Copy_ProductTable_To_CSV
- Ingestion_Customer_SQLDB_ADLS
- Ingestion_Customer_SQLDB_ADLS_F...
- Assignment_5_5_Ingestion_Product...
- Assignment_5_5_ChildPipeline_2
- Ingestion_Product_To_JSON
- Change Data Capture (preview) 0
- Datasets 11**
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_Generic_JSON_DS
 - ADLS_JSON_Dataset
 - SQL_Address_Table_DS
 - SQL_Customer_JOIN_CustomerAddr...
 - SQL_Customer_Table_Dataset

Activities

- Copy
- Move and transform
 - Copy data

Diagram: A pipeline diagram showing a 'Lookup' activity 'Get Product record count' leading to an 'If Condition' activity 'Check record count gt 10'. The 'True' branch leads to a 'Copy product...' activity.

Pipeline run ID: 21049ce8-2132-4a37-bcf2-51c6cf56c644 **Pipeline status:** Succeeded

Activity log:

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties
Copy product records as JSON	Succeeded	Copy data	2/7/2024, 12:35:53 AM	13s	AutoResolveIntegration	
Check record count gt 10	Succeeded	If Condition	2/7/2024, 12:35:52 AM	15s		
Get Product record count	Succeeded	Lookup	2/7/2024, 12:35:46 AM	5s	AutoResolveIntegration	

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Factory Resources

- Assignment_5_4_Ingestion_Product...
- Copy_Customer_JSON_To_Folder
- Copy_Customer_Table_To_CSV
- Copy_Customer_Table_To_CSV_Pipe
- Copy_Customer_To_JSON
- Copy_ProductTable_To_CSV
- Ingestion_Customer_SQLDB_ADLS
- Ingestion_Customer_SQLDB_ADLS_F...
- Assignment_5_5_Ingestion_Product...
- Assignment_5_5_ChildPipeline...
- Ingestion_Product_To_JSON
- Change Data Capture (preview) 0
- Datasets 11**
 - ADLS_Customer_CSV_Dataset
 - ADLS_Customer_CSV_Pipe
 - ADLS_Datalake_storage_CSV_Dataset
 - ADLS_Generic_CSV_DS
 - ADLS_Generic_JSON_DS
 - ADLS_JSON_Dataset
 - SQL_Address_Table_DS
 - SQL_Customer_JOIN_CustomerAddr...
 - SQL_Customer_Table_Dataset

Activities

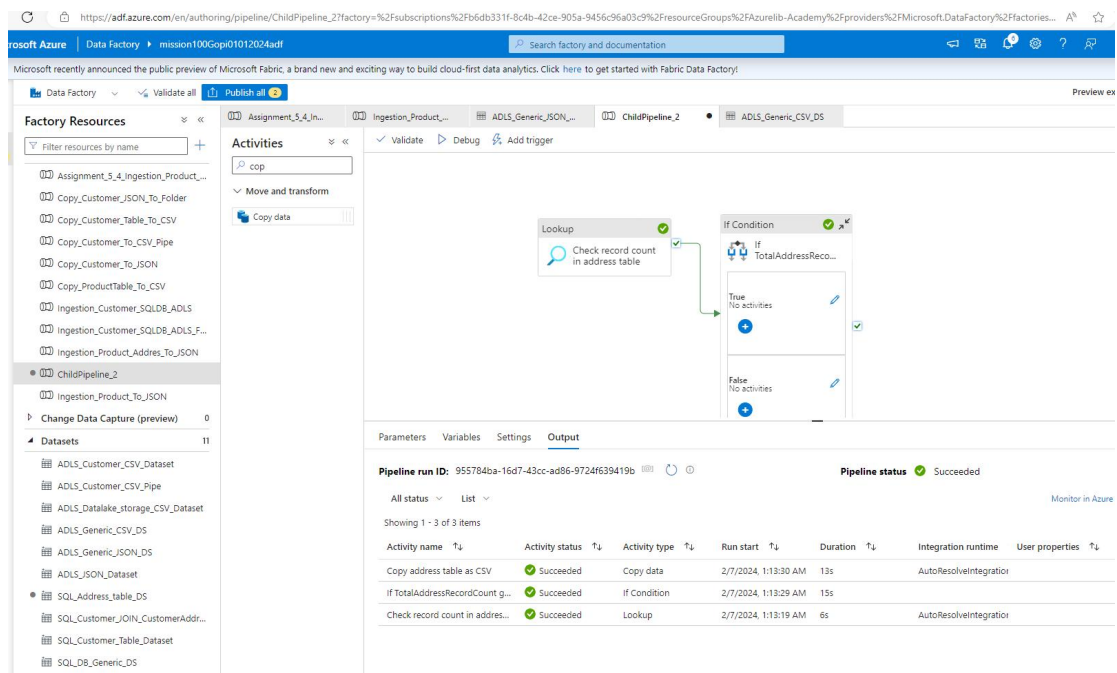
- Copy
- Move and transform
 - Copy data

Diagram: A pipeline diagram showing a 'Lookup' activity 'Check record count in address table' leading to an 'If Condition' activity 'If TotalAddressRecordCount g...'. The 'True' branch leads to a 'Copy data' activity.

Pipeline run ID: 955784ba-16d7-43cc-ad86-9724f639419b **Pipeline status:** Succeeded

Activity log:

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User pr
Copy address table as CSV	Succeeded	Copy data	2/7/2024, 1:13:30 AM	13s	AutoResolveIntegration	
If TotalAddressRecordCount g...	Succeeded	If Condition	2/7/2024, 1:13:29 AM	15s		
Check record count in addres...	Succeeded	Lookup	2/7/2024, 1:13:19 AM	6s	AutoResolveIntegration	



Microsoft Azure | Data Factory | mission100Gop1012024adf

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Factory Resources

Activities

Lookup: Check record count in address table

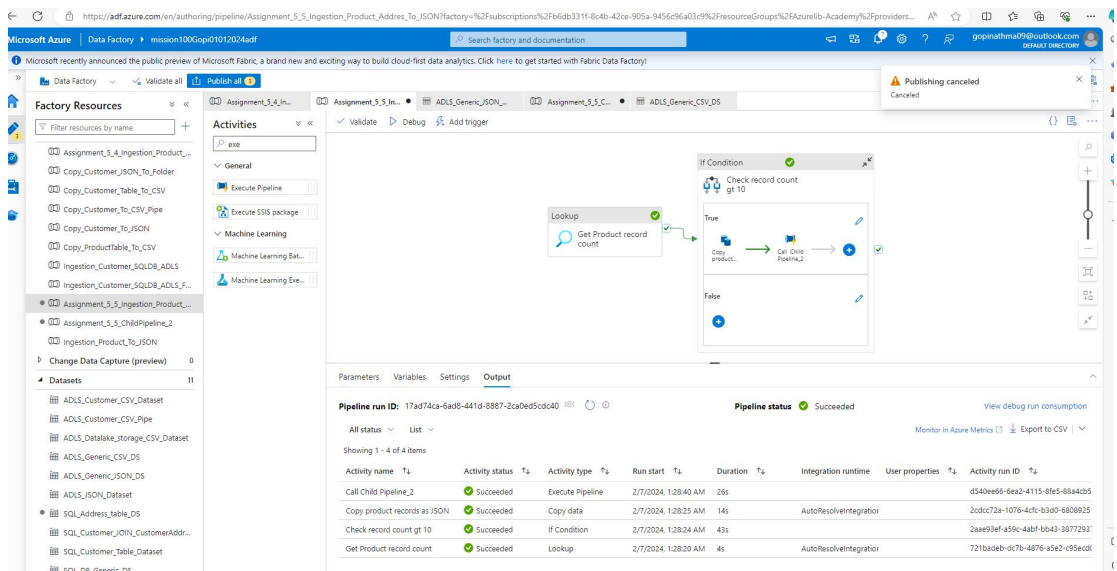
If Condition: If TotalAddressReco...

Copy data: Copy data

Pipeline run ID: 955784ba-16d7-43cc-ad86-9724f639419b

Pipeline status: Succeeded

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties
Copy address table as CSV	Succeeded	Copy data	2/7/2024, 1:13:30 AM	13s	AutoResolveIntegration	
If totalAddressRecordCount g...	Succeeded	If Condition	2/7/2024, 1:13:29 AM	15s	AutoResolveIntegration	
Check record count in address...	Succeeded	Lookup	2/7/2024, 1:13:19 AM	6s	AutoResolveIntegration	



Microsoft Azure | Data Factory | mission100Gop1012024adf

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Factory Resources

Activities

Lookup: Get Product record count

If Condition: Check record count gt 10

Copy product...: Copy product...

Call Child Pipeline_2: Call Child Pipeline_2

Pipeline run ID: 17ad74ca-6ad8-441d-8887-2ca0ed5cd40c

Pipeline status: Succeeded

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Call Child Pipeline_2	Succeeded	Execute Pipeline	2/7/2024, 1:28:40 AM	26s			d540ee66-6ea2-4115-8f65-88a4cb5
Copy product records as JSON	Succeeded	Copy data	2/7/2024, 1:28:25 AM	14s	AutoResolveIntegration		2cdc72a-1078-4cfc-b3d0-8808925
Check record count gt 10	Succeeded	If Condition	2/7/2024, 1:28:24 AM	43s	AutoResolveIntegration		2aae93ef-a59c-4abf-bb49-3877293
Get Product record count	Succeeded	Lookup	2/7/2024, 1:28:20 AM	4s	AutoResolveIntegration		721ba0eb-dc7b-4878-a5e2-c95e0d8

Assignment 6: ADF Pipeline 3

Azurelib Academy

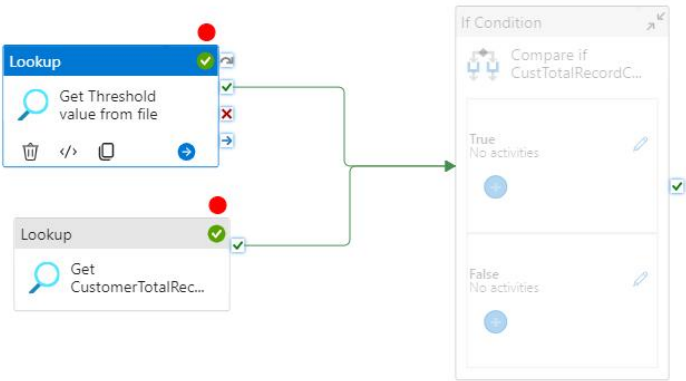
6.I am attaching a file, load this file into your ADLS location. This file contains a threshold value.

Create a pipeline

in such a manner that, it will first get the threshold value from this file and then check if the record count in the customer

table is more than it or not. If yes then copy the customer data from SQL db to ADLS location in JSON format.

Solution:



General **Settings** User properties

Source dataset * Threshold_text_file_DS [Open](#) [+ New](#) [Preview data](#) [Learn more](#)

First row only ☐

File path type ☒ File path in dataset ☐ Wildcard file path ☐ List of files

Filter by last modified

Start time (UTC) End time (UTC)

Recursively ☒

Enable partitions discovery ☐

Max concurrent connections

Skip line count

Preview data

Linked service: ADLS_Azure_DataLakeStorage1_Linked_service

Object: Threshold_file.txt

	Prop_0
1	250

Assignment_6_part...

Validate Cancel options Add trigger

Lookup
Get Threshold value from file

Lookup
Get CustomerTotalRec...

If Condition
Compare if CustTotalRecordC...

True
No activities

False
No activities

Output

Copy to clipboard

```
{
  "count": 1,
  "value": [
    {
      "Prop_0": "250"
    }
  ],
  "effectiveIntegrationRuntime":
    "AutoResolveIntegrationRuntime (East US)",
}
```

	type	Run start	Duration
Get Threshold value...	Lookup	2/7/2024, 9:21:07 PM	3s
Get CustomerTotalRecordCou...	Lookup	2/7/2024, 9:21:07 PM	5s

```
int(activity('Get Threshold value from file').output.value[0].Prop_0)
```

Assignment_o_part...

Validate Debug Add trigger

Lookup

Get Threshold value from file

Lookup

Get CustomerTotalRec...

If Condition

Compare if CustTotalRecordC...

True No activities

False No activities

Output

Copy to clipboard

```
{
  "firstRow": {
    "CustomerTotalRecordCount": 847
  },
  "effectiveIntegrationRuntime":
  "AutoResolveIntegrationRuntime (East US)",
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
```

	type	Run start	Durati
Get CustomerTotalRecordCou...	Succeeded	2/7/2024, 9:21:46 PM	4s
Lookup		2/7/2024, 9:21:46 PM	3s

DETAILS DIRECTORY

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@greater(activity('Get CustomerTotalRecordCount').output.firstRow.CustomerTotalRecordCount, int(activity('Get Threshold value from file').output.value[0].Prop_0))
```

Clear contents

Assignment_6_part...

« ✓ Validate ✓ Validate copy runtime ▶ Debug ⚙ Add trigger

Assignment_6_part6_threshold_file > ⚙ Compare if CustTotalRecordCount gt thresholdvalue > True activities

Copy data

Copy Customer data to ADLS JSON

General Source Sink Mapping Settings User properties

Source dataset * SQL_Customer_Table_Dataset Open New Preview data [Learn more](#)

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Partition option ☒ None ☐ Physical partitions of table ☐ Dynamic range

Please preview data to validate the partition settings.

Additional columns New

Assignment_6_part...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Assignment_6_part6_threshold_file > ⚡ Compare if CustTotalRecordCount gt thresholdvalue > True activities

Copy data

Copy Customer data to ADLS JSON

🗑️ </> 📄 ➕ ➡

General Source Sink Mapping Settings User properties

Sink dataset *

ADLS_Generic_JSON_DS

Open + New Learn more

Dataset properties

Name	Value
folderName_param	Assignment_6_6_Threshold_Customer_JS...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

+ New

File pattern

2024ads | Containers > landing >

Upload + Add Directory

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: landing / Assignment_6_6_Threshold_Customer_JSON_OUPUT
Search blobs by prefix (case-...)
☐ Show deleted objects

Name

☐ {}

☒ SalesLT.Customer.json

Assignment_6_6_Threshold_Customer_JSON_OUPUT/SalesLT.Customer.json

Save ✕ Discard ↓ Download ↻ Refresh 🗑️ Delete

Overview Versions Edit Generate SAS

1 { "CustomerID":1, "NameStyle":false, "Title":"Mr.", "FirstName":"Orlando", "MiddleName":"M.", "LastName":"Gee", "Suffix":null, "CompanyName":"A Bike Store", "Sale

2 { "CustomerID":2, "NameStyle":false, "Title":"Mr.", "FirstName":"Keith", "MiddleName":null, "LastName":"Harris", "Suffix":null, "CompanyName":"Progressive Sports

3 { "CustomerID":3, "NameStyle":false, "Title":"Ms.", "FirstName":"Donna", "MiddleName":"F.", "LastName":"Carreras", "Suffix":null, "CompanyName":"Advanced Bike Co

4 { "CustomerID":4, "NameStyle":false, "Title":"Ms.", "FirstName":"Janet", "MiddleName":"M.", "LastName":"Gates", "Suffix":null, "CompanyName":"Metropolitan Cycle Syste

5 { "CustomerID":5, "NameStyle":false, "Title":"Mr.", "FirstName":"Lucy", "MiddleName":null, "LastName":"Harrington", "Suffix":null, "CompanyName":"Metropolitan Sp

6 { "CustomerID":6, "NameStyle":false, "Title":"Ms.", "FirstName":"Rosemarie", "MiddleName":"J.", "LastName":"Carroll", "Suffix":null, "CompanyName":"Aerobic Exercis

7 { "CustomerID":7, "NameStyle":false, "Title":"Mr.", "FirstName":"Dominic", "MiddleName":"P.", "LastName":"Gash", "Suffix":null, "CompanyName":"Associated Bikes",

8 { "CustomerID":8, "NameStyle":false, "Title":"Ms.", "FirstName":"Kathleen", "MiddleName":"M.", "LastName":"Garcia", "Suffix":null, "CompanyName":"Rural Cycle Exp

9 { "CustomerID":9, "NameStyle":false, "Title":"Ms.", "FirstName":"Katherine", "MiddleName":null, "LastName":"Harding", "Suffix":null, "CompanyName":"Sharp Bikes

10 { "CustomerID":10, "NameStyle":false, "Title":"Mr.", "FirstName":"Johnny", "MiddleName":"A.", "LastName":"Caprio", "Suffix":null, "CompanyName":"Bikes and Motor

11 { "CustomerID":11, "NameStyle":false, "Title":"Mr.", "FirstName":"Christopher", "MiddleName":"R.", "LastName":"Beck", "Suffix":null, "CompanyName":"Bulk Discoun

12 { "CustomerID":12, "NameStyle":false, "Title":"Mr.", "FirstName":"David", "MiddleName":"J.", "LastName":"Liu", "Suffix":null, "CompanyName":"Catalog Store", "Sale

13 { "CustomerID":13, "NameStyle":false, "Title":"Mr.", "FirstName":"John", "MiddleName":"A.", "LastName":"Beaver", "Suffix":null, "CompanyName":"Center Cycle Shop

14 { "CustomerID":14, "NameStyle":false, "Title":"Ms.", "FirstName":"Jean", "MiddleName":"P.", "LastName":"Handley", "Suffix":null, "CompanyName":"Central Discoun

15 { "CustomerID":15, "NameStyle":false, "Title":null, "FirstName":"Linghao", "MiddleName":null, "LastName":"Liu", "Suffix":null, "CompanyName":"Chic Department Sto

16 { "CustomerID":16, "NameStyle":false, "Title":"Ms.", "FirstName":"Linda", "MiddleName":"E.", "LastName":"Burnett", "Suffix":null, "CompanyName":"Travel Systems",

17 { "CustomerID":17, "NameStyle":false, "Title":"Mr.", "FirstName":"Kerim", "MiddleName":null, "LastName":"Hanif", "Suffix":null, "CompanyName":"Bike World", "Sales

18 { "CustomerID":18, "NameStyle":false, "Title":"Mr.", "FirstName":"Kevin", "MiddleName":null, "LastName":"Liu", "Suffix":null, "CompanyName":"Eastside Department

19 { "CustomerID":19, "NameStyle":false, "Title":"Mr.", "FirstName":"Donald", "MiddleName":"E.", "LastName":"Blanton", "Suffix":null, "CompanyName":"Coalition Bike

20 { "CustomerID":20, "NameStyle":false, "Title":"Mr.", "FirstName":"Jackie", "MiddleName":"E.", "LastName":"Blackwell", "Suffix":null, "CompanyName":"Commuter Bicy

21 { "CustomerID":21, "NameStyle":false, "Title":"Mr.", "FirstName":"Bryan", "MiddleName":null, "LastName":"Hamilton", "Suffix":null, "CompanyName":"Cross-Country R

22 { "CustomerID":22, "NameStyle":false, "Title":"Mr.", "FirstName":"Todd", "MiddleName":"R.", "LastName":"Logan", "Suffix":null, "CompanyName":"Cycle Merchants", "S

23 { "CustomerID":23, "NameStyle":false, "Title":"Ms.", "FirstName":"Barbara", "MiddleName":"J.", "LastName":"German", "Suffix":null, "CompanyName":"Cycles Wholesale

24 { "CustomerID":24, "NameStyle":false, "Title":"Mr.", "FirstName":"Jim", "MiddleName":null, "LastName":"Geist", "Suffix":null, "CompanyName":"Two Bike Shops", "Sal

25 { "CustomerID":25, "NameStyle":false, "Title":"Ms.", "FirstName":"Betty", "MiddleName":"M.", "LastName":"Haines", "Suffix":null, "CompanyName":"Finer Mart", "Sale

26 { "CustomerID":26, "NameStyle":false, "Title":"Ms.", "FirstName":"Sharon", "MiddleName":"J.", "LastName":"Looney", "Suffix":null, "CompanyName":"Journey Hotel", "S

27 { "CustomerID":27, "NameStyle":false, "Title":"Mr.", "FirstName":"Darren", "MiddleName":null, "LastName":"Gehring", "Suffix":null, "CompanyName":"Journey Sportin

28 { "CustomerID":28, "NameStyle":false, "Title":"Ms.", "FirstName":"Erin", "MiddleName":"M.", "LastName":"Hagens", "Suffix":null, "CompanyName":"Distant Inn", "Sale

29 { "CustomerID":29, "NameStyle":false, "Title":"Mr.", "FirstName":"Jeremy", "MiddleName":null, "LastName":"Lus", "Suffix":null, "CompanyName":"Wealthy Activity St

30 { "CustomerID":30, "NameStyle":false, "Title":"Ms.", "FirstName":"Elsa", "MiddleName":null, "LastName":"Leavitt", "Suffix":null, "CompanyName":"Frugal Bike Shop"

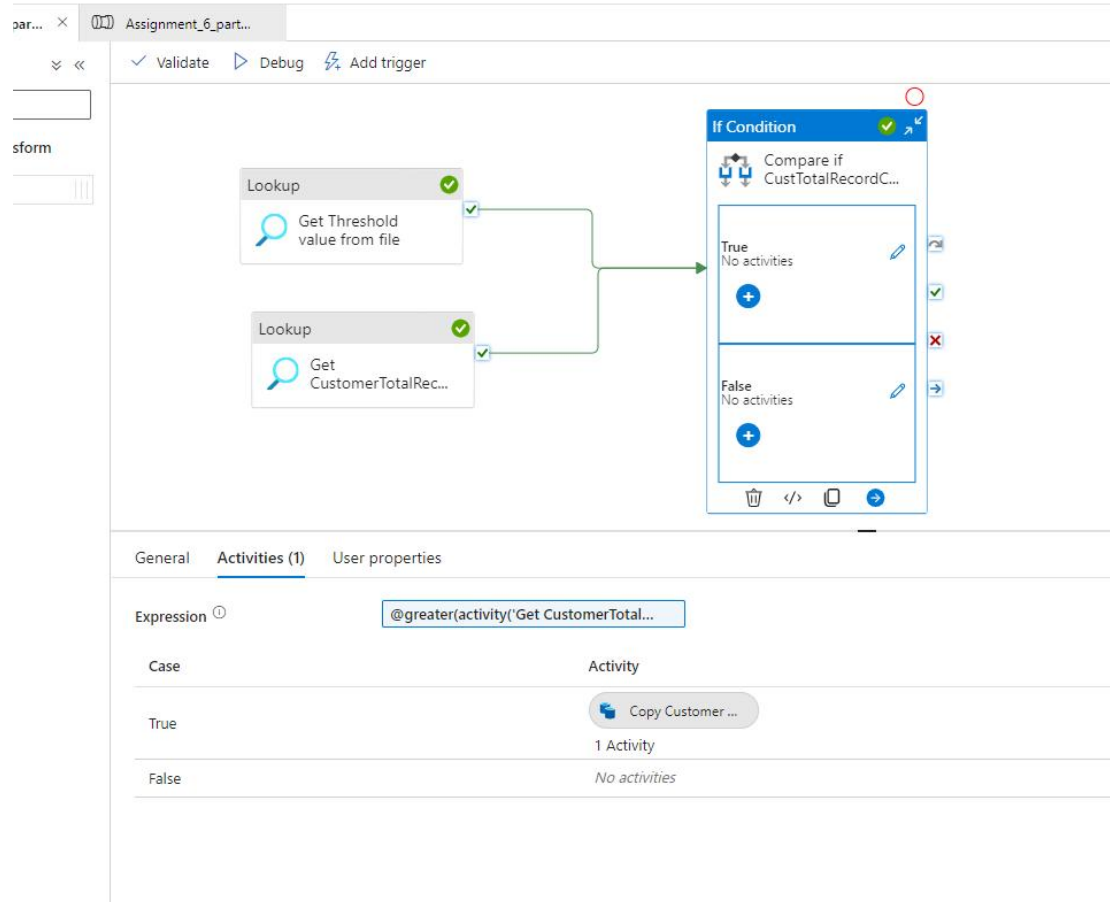
31 { "CustomerID":31, "NameStyle":false, "Title":"Mr.", "FirstName":"David", "MiddleName":"O.", "LastName":"Lawrence", "Suffix":null, "CompanyName":"Gear-Shift Bikes

32 { "CustomerID":32, "NameStyle":false, "Title":"Ms.", "FirstName":"Hattie", "MiddleName":"J.", "LastName":"Haemon", "Suffix":null, "CompanyName":"Greater Bike Sto

json

Preview

and new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!



5. In the above case problem is that every time the pipeline runs it will overwrite the data. Ensure that all the data goes into Folder like (Customer/Year/Month/Day) to avoid any overwriting. Please rewrite the pipeline

Assignment 6: ADF Pipeline 3

[Azurelib Academy](#)

Easy Questions:

1. Create a pipeline name 'Without_Foreach_Example' to copy the Customer & Customer address table data into ADLS without using foreach activity.

Solution:

Use 2 copy activity : 1 for Customer dAta, 2nd for Customer Address and make connection

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines (13)
 - Assignment_5_4_Ingestion_Product_To_JSON
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product_Addres_To_JSON
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
 - Without_Foreach_Example_Pipeline**
 - Change Data Capture (preview) 0
- Datasets (12)
- Data flows (0)
- Power Query (0)

Activities

Assignment_6_part... Without_Foreach_...

Validate Validate copy runtime Debug Add trigger

Copy data

Copy data

Copy Customer Data

Copy Customer Address

General Source Sink Mapping Settings User properties

Source dataset SQL_DB_Generic_DS Open New Preview data Learn more

Dataset properties

Name	Value
schemaName	SalesLT
tableName	CustomerAddress

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines (13)
 - Assignment_5_4_Ingestion_Product_To_JSON
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product_Addres_To_JSON
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
 - Without_Foreach_Example_Pipeline**
 - Change Data Capture (preview) 0
- Datasets (12)
- Data flows (0)
- Power Query (0)

Activities

Assignment_6_part... Without_Foreach_...

Validate Validate copy runtime Debug Add trigger

Copy data

Copy data

Copy Customer Data

Copy Customer Address

General Source Sink Mapping Settings User properties

Name Copy Customer Data Learn more

Description Copy Customer data to ADLS

Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0

Retry interval (sec) 30

Without_Foreach_...

Validate

Validate copy runtime

Debug

Add trigger

Copy data

Copy Customer Data

</>

Copy data

Copy Customer Address

General

Source

Sink

Mapping

Settings

User properties

Source dataset *

SQL_DB_Generic_DS

Open

New

Preview data

Learn more

Dataset properties

Name	Value
schemaName	SalesLt
tableName	Customer

Use query

Table

Query

Stored procedure

Query timeout (minutes)

120

Isolation level

Select...

Partition option

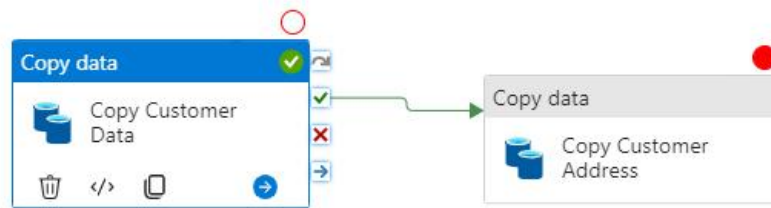
None

Physical partitions of table

Dynamic range

Without_Foreach_...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger



General Source **Sink** Mapping Settings User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open + New Learn more

Dataset properties ⓘ

Name

Value

folderName_param

Assignment6_part_1_Customer_without_...

Copy behavior ⓘ

Select...

Max concurrent connections ⓘ

Block size (MB) ⓘ

Metadata ⓘ

+ New

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 13
 - Assignment_5_4_Ingestion_Product_To_JSON
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product_Addres_To_JSON
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
 - Without_Foreach_Example_Pipeline**
 - Change Data Capture (preview) 0
 - Datasets 12
 - Data flows 0
 - Power Query 0

Activities

Assignment_6_part... Without_Foreach_...

Validate Validate copy runtime Debug Add trigger

Move and transform

- Copy data
- Azure Data Explorer
 - Azure Data Explorer C...
- Azure Function
 - Azure Function
- Batch Service
 - Custom
- Databricks
 - Notebook
 - Jar
- Python
- Data Lake Analytics
 - U-SQL
- General
 - Execute Pipeline
 - Execute SSIS package
 - Stored procedure

Copy data

Copy Customer Data → Copy Customer Address

General **Source** Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS Open + New Preview data Learn more

Dataset properties

Name	Value
schemaName	SalesLT
tableName	CustomerAddress

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Without_Foreach_...

Validate Validate copy runtime Debug Add trigger

Copy data

Copy Customer Data → Copy Customer Address

General **Sink** Mapping Settings User properties

Sink dataset * ADLS_Generic_CSV_DS Open + New Learn more

Dataset properties

Name	Value
folderName_param	Assignment6_part1_CustomerAddress_C...

Copy behavior Select...

Max concurrent connections

Block size (MB)

Metadata + New

je accounts > gopi01012024adls | Containers >

ng ...

Upload

Add Directory

Refresh

Rename

Delete

Change tier

Acquire lease

Break lease

Give feedback

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	Blk
☐ Address				
☐ Assignment_5_Address_Table_CSV				
☐ Assignment_5_Copy_Product_Table_JSON				
☐ Assignment6_part_1_Customer_without_Foreach				
☐ Assignment6_part1_CustomerAddress_CSV_without_Foreach				
☐ CSV				
☐ CSV_Pipe				
☐ CSV2				
☐ JSON				
☐ Practice-1				
☐ Product_JSON				
☐ Threshold_file.txt	2/7/2024, 9:08:32 PM	Hot (Inferred)		Blk
☐ Threshold-TwvFV4QAZU-FUIXd8kxvf.txt	2/7/2024, 8:14:54 PM	Hot (Inferred)		Blk

Microsoft Azure Upgrade

Search resources, services, and docs (G+)

Home > Storage accounts > gopi01012024adls | Containers > landing >

landing

Container

Search

Upload

Add Directory

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / Assignment6_part_1_Customer_without_Foreach

Search blobs by prefix (case-sensitive)

Show deleted objects

Name	Modified	Access tier	Archive status	Blk
☐ I-I				
☒ SalesLt.Customer.txt				

Assignment6_part_1_Customer_without_Foreach/SalesLt.Customer.txt

Save

Discard

Download

Refresh

Delete

Overview

Versions

Edit

Generate SAS

```
1 CustomerID,NameStyle,Title,FirstName,MiddleName,LastName,Suffix,CompanyName,SalesPerson,EmailAddress,Phone,PasswordHash,PasswordSalt
2 1,False,"Mr.", "Orlando", "N.", "Gee", "A Bike Store", "adventure-works\pamelao", "orlando@adventure-works.com", "245-555-0173", "L/Rluczp
3 2,False,"Mr.", "Keith", "Harris", "Progressive Sports", "adventure-works\david08", "keith@adventure-works.com", "178-555-0127", "VPdRdqv
4 3,False,"Ms.", "Donna", "F.", "Carrenas", "Advanced Bike Components", "adventure-works\jillian0", "donna@adventure-works.com", "279-555-0
5 4,False,"Ms.", "Janet", "M.", "Gates", "Modular Cycle Systems", "adventure-works\jillian0", "janet@adventure-works.com", "718-555-0173",
6 5,False,"Ms.", "Lucy", "Harrington", "Metropolitan Sports Supply", "adventure-works\shu0", "lucy@adventure-works.com", "828-555-0186",
7 6,False,"Ms.", "Rosmarie", "J.", "Carroll", "Aerobic Exercise Company", "adventure-works\linda3", "rosmarie@adventure-works.com", "244-55
8 7,False,"Mr.", "Dominic", "P.", "Gash", "Associated Bikes", "adventure-works\shu0", "dominl@adventure-works.com", "192-555-0173", "ZcooP/
9 10,False,"Ms.", "Kathleen", "H.", "Garza", "Rural Cycle Emporium", "adventure-works\josel1", "kathleen@adventure-works.com", "158-555-0127
10 11,False,"Ms.", "Katherine", "H.", "Harding", "Sharp Bikes", "adventure-works\josel1", "katherine@adventure-works.com", "926-555-0159", "uRloV/
11 12,False,"Mr.", "Johnny", "A.", "Caprio", "Bikes and Motorbikes", "adventure-works\garrett1", "johnny@adventure-works.com", "112-555-
12 16,False,"Mr.", "Christopher", "R.", "Beck", "Bulk Discount Store", "adventure-works\jaeo", "christopher@adventure-works.com", "1 (1
13 18,False,"Mr.", "David", "J.", "Liu", "Catalog Store", "adventure-works\michael19", "david02@adventure-works.com", "448-555-0132", "61zeIK0+
14 19,False,"Mr.", "John", "A.", "Beaver", "Center Cycle Shop", "adventure-works\pamelao", "john@adventure-works.com", "521-555-0195", "DibQd
15 20,False,"Ms.", "Jean", "P.", "Handley", "Central Discount Store", "adventure-works\david08", "jean1@adventure-works.com", "582-555-0113",
16 21,False,"Jinghao", "Liu", "Chic Department Stores", "adventure-works\jillian0", "jinghao1@adventure-works.com", "928-555-0116", "TaD5A
17 22,False,"Ms.", "Linda", "E.", "Burnett", "Travel Systems", "adventure-works\jillian0", "linda4@adventure-works.com", "121-555-0121", "23Aw
18 23,False,"Mr.", "Kerla", "Harris", "Bike World", "adventure-works\shu0", "kerla@adventure-works.com", "216-555-0122", "58N5JosAdP73K0W/N4
19 24,False,"Mr.", "Kevin", "Liu", "Eastside Department Store", "adventure-works\linda3", "kevin5@adventure-works.com", "926-555-0164", "yIT
20 25,False,"Mr.", "Donald", "J.", "Blanton", "Coalition Bike Company", "adventure-works\shu0", "donald0@adventure-works.com", "357-555-0161"
21 28,False,"Mr.", "Jackie", "E.", "Blackwell", "Commuter Bicycle Store", "adventure-works\josel1", "jackie@adventure-works.com", "972-555-01
22 29,False,"Mr.", "Bryan", "Hamilton", "Cross-Country Riding Supplies", "adventure-works\josel1", "bryan@adventure-works.com", "344-555-01
23 30,False,"Mr.", "Todd", "R.", "Logan", "Cycle Merchants", "adventure-works\garrett1", "todd@adventure-works.com", "783-555-0110", "7V6c8Dy
24 34,False,"Ms.", "Barbara", "J.", "German", "Cycles Wholesaler & Hfg.", "adventure-works\jaeo", "barbara4@adventure-works.com", "1 (11) 580
25 37,False,"Mr.", "Jim", "Gelst", "Two Bike Shops", "adventure-works\pamelao", "jini@adventure-works.com", "724-555-0161", "CvqeC4f1CkUj9J1
26 38,False,"Ms.", "Betty", "H.", "Haines", "Finer Hart", "adventure-works\david08", "betty@adventure-works.com", "867-555-0114", "Q/n6Wz20012
27 39,False,"Ms.", "Sharon", "J.", "Looney", "Fitness Hotel", "adventure-works\jillian0", "sharon2@adventure-works.com", "377-555-0132", "Uo3K
28 40,False,"Mr.", "Darren", "J.", "Hegring", "Journey Sporting Goods", "adventure-works\jillian0", "darren@adventure-works.com", "417-555-0182"
29 41,False,"Ms.", "Erin", "H.", "Hagens", "Distant Inn", "adventure-works\shu0", "erini@adventure-works.com", "244-555-0127", "925fu/b18d0U09
30 42,False,"Mr.", "Jeremy", "J.", "Los", "Healthy Activity Store", "adventure-works\linda3", "jeremy@adventure-works.com", "911-555-0165", "JULK
31 43,False,"Ms.", "Elsa", "Leavitt", "Frugal Bike Shop", "adventure-works\shu0", "elsa@adventure-works.com", "482-555-0174", "Bm3JHw1476Nk
32 46,False,"Mr.", "David", "O.", "Lawrence", "Gear-Shift Bikes Limited", "adventure-works\josel1", "david19@adventure-works.com", "653-555-015
```

Microsoft Azure portal interface showing the 'Assignment6_part1_CustomerAddress_CSV_without_Foreach/SalesLT.CustomerAddress.txt' blob. The interface includes a left sidebar with navigation options like 'Storage accounts', 'Containers', and 'landing'. The main content area displays the blob's details, including its name, location, and a list of rows. The 'Overview' tab is selected, showing a table of data with columns: CustomerID, AddressID, AddressType, rowguid, ModifiedDate. The table contains 32 rows of data, each representing a customer address record.

2. Create a pipeline name 'Foreach Example' to copy the Customer & Customer address table data into ADLS using the one copy activity.

Solution:

We use “For Each” Activity ++ Copy Activity

Microsoft Fabric Data Factory interface showing the 'Assignment6_Foreach_Example' pipeline. The pipeline is configured with a 'For Each' activity (Foreach1) that iterates over a dataset and executes a 'Copy data' activity. The 'Copy data' activity is configured to copy data from the 'SalesLT.CustomerAddress' table in the 'AdventureworksLT' database to the 'Assignment6_part1_CustomerAddress_CSV_without_Foreach' folder in the 'SalesLT' storage account. The pipeline is currently in the 'Activated' state.

Factory Resources

Filter resources by name

Pipelines 14

- Assignment_5_4_Ingestion_Product_To_JSON
- Assignment_5_5_ChildPipeline_2
- Assignment_5_5_Ingestion_Product_Addres_To_JSON
- Assignment_6_part6_threshold_file
- Copy_Customer_JSON_To_Folder
- Copy_Customer_Table_To_CSV
- Copy_Customer_To_CSV_Pipe
- Copy_Customer_To_JSON
- Copy_ProductTable_To_CSV
- Ingestion_Customer_SQLDB_ADLS
- Ingestion_Customer_SQLDB_ADLS_Folder
- Ingestion_Product_To_JSON
- Assignment_6_Foreach_Example
- Without_Foreach_Example_Pipeline

Change Data Capture (preview) 0

Datasets 12

Data flows 0

Power Query 0

Activities

for

Move and transform

- Copy data
- Data flow

Iteration & conditionals

- ForEach

ForEach

ForEach1

Activities

No activities

Settings

General

Sequential ☐

Batch count

Items *

This property should be parameterized.

Add dynamic content [Alt+Shift+D]

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@createArray()
```

❌ function 'createArray' does not accept 0 argument(s)

[Clear contents](#)

Parameters

System variables

Functions

Variables

 create

 [Collapse all](#)

▼ **Conversion Functions**

createArray

Creates an array from the parameters. For example, the following expression returns ["a", "c"]: cr...

gopinathmadas@outlook.com

DEFAULT DIRECTORY

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

@createArray('Customer','CustomerAddress')

[Clear contents](#)

Parameters

System variables

Functions

Variables

create

^ Collapse all

▼ Conversion Functions

createArray

Creates an array from the parameters. For example, the following expression returns ["a", "c"]: creat...

ntly announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

actory

Validate all

Publish all

Preview experience

Resources

sources by name

es

14

ignment_5_4_Ingestion_Product_To_JSON

ignment_5_5_ChildPipeline_2

ignment_5_5_Ingestion_Product_Addres_To_JSON

ignment_6_part6_threshold_file

ry_Customer_JSON_To_Folder

ry_Customer_Table_To_CSV

ry_Customer_To_CSV_Pipe

ry_Customer_To_JSON

ry_ProductTable_To_CSV

estion_Customer_SQLDB_ADLS

estion_Customer_SQLDB_ADLS_Folder

estion_Product_To_JSON

ignemnt_6_Foreach_Example

hout_Foreach_Example_Pipeline

> Data Capture (preview)

0

ts

12

pws

0

-

Activities

for

Move and transform

Copy data

Data flow

Iteration & conditionals

ForEach

ForEach

ForEach1

Activities

No activities

General

Settings

Activities (0)

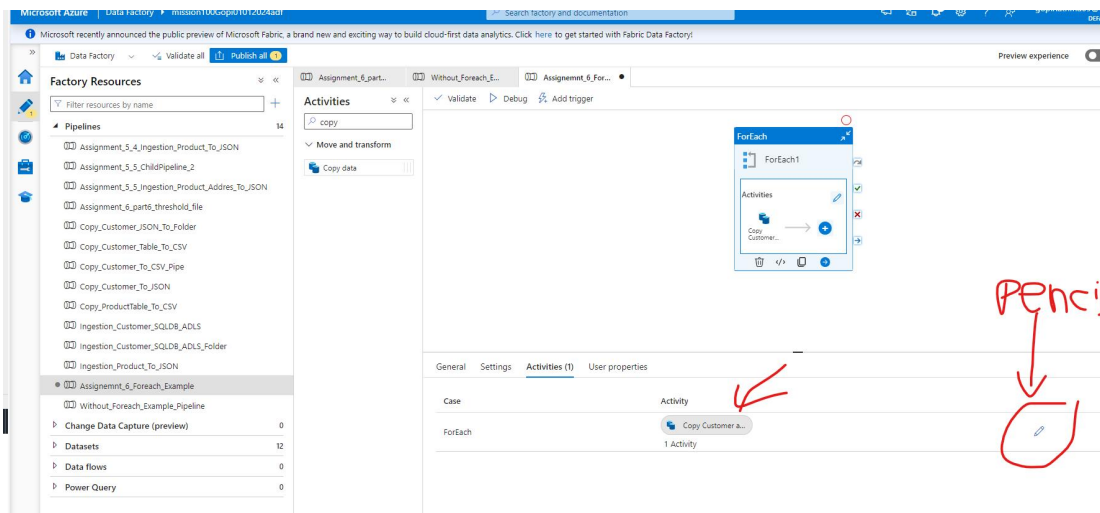
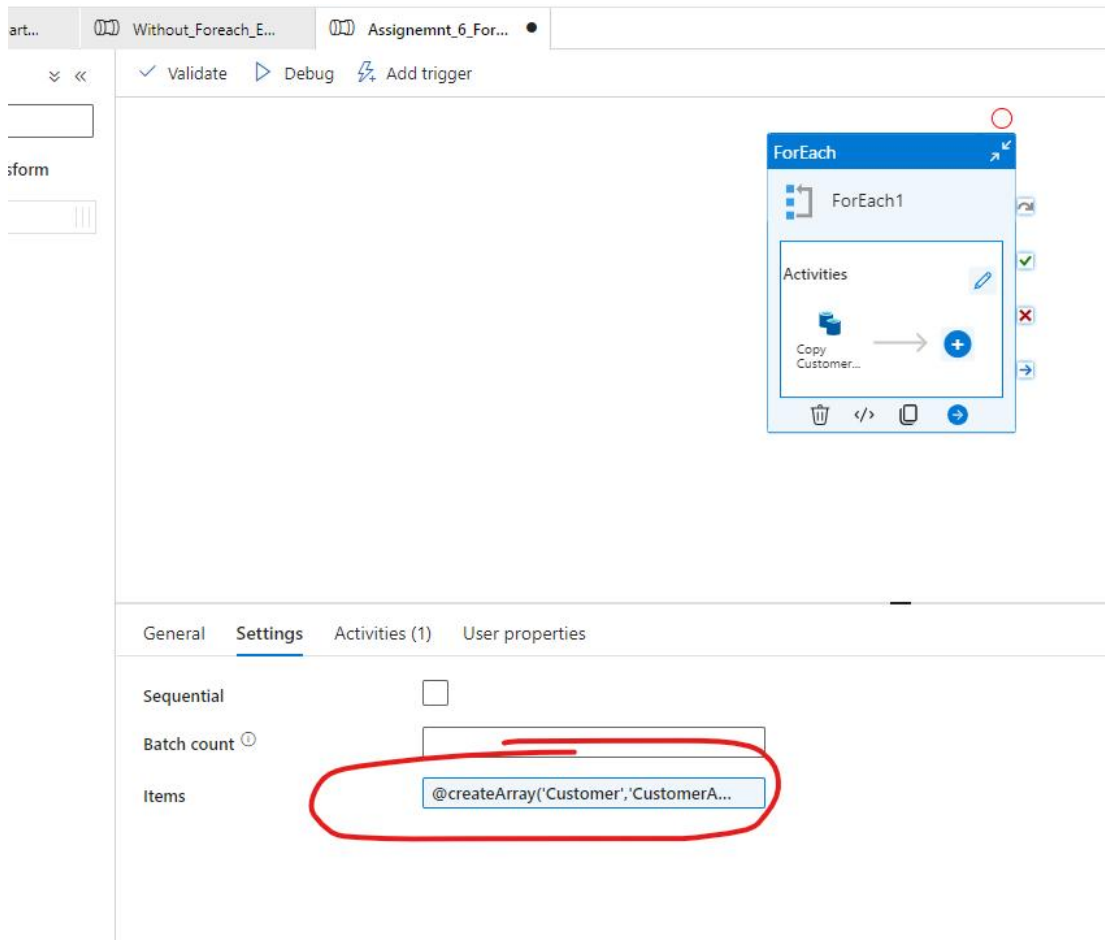
User properties

Case

Activity

ForEach

No activities



Assignment_6_part... Without_Foreach_E... Assignemnt_6_For...

Activities

copy

Move and transform

Copy data

ForEach

ForEach1

Activities

Copy Customer...

General Settings Activities (1) User properties

Case

ForEach

Activity

Copy Customer a...

1 Activity

Assignment_6_part... Without_Foreach_E... Assignemnt_6_For...

Activities

copy

Move and transform

Copy data

ForEach

ForEach1

Activities

Copy Customer...

General Settings Activities (1) User properties

Case

ForEach

Activity

Copy Customer a...

1 Activity

brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Assignment_6_part... Without_Foreach_E... Assignment_6_For...

Activities

- copy
- Move and transform
 - Copy data

Assignment_6_Foreach_Example > ForEach1

Copy data

Copy Customer and Cusotmer...

General Source Sink Mapping Settings User properties

Name * Copy Customer and Customer Address [Learn more](#)

Description 'Foreach Example' to copy the Customer & Customer address table data into ADLS using the one copy activity

Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0

Retry interval (sec) 30

Analytics. Click [here](#) to get started with Fabric Data Factory!

Assignemnt_6_For...

Validate copy runtime Debug Add trigger

_6_Foreach_Example > ForEach1 runs 2times

SQL_DB_Generic_DS

Dataset properties

Name	Value
schemaName	SalesLT
tableName	@item()

Table Query Stored procedure

minutes) 120

Select...

None Physical partitions of table Dynamic

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#)

@item()

Clear contents

ForEach iterator

Parameters

System variable

Search

ForEach1 runs 2times
Current item

OK

Cancel

Without_Foreach_E... Assignemnt_6_For...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Assignemnt_6_Foreach_Example > ForEach1 runs 2times

Copy Customer and Cusotmer...

General **Source** Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS Open + New Preview data Learn more

Dataset properties

Name	Value
schemaName	SalesLT
tableName	@item()

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Partition option ☒ None ☐ Physical partitions of table ☐ Dynamic range

Without_Foreach_E...

Assignemnt_6_For...

Validate

Validate copy runtime

Debug

Add trigger

Assignemnt_6_Foreach_Example > ForEach1

Copy Customer and Cusotmer...

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name	Value
folderName_param	@item()

Copy behavior

Select...

Max concurrent connections

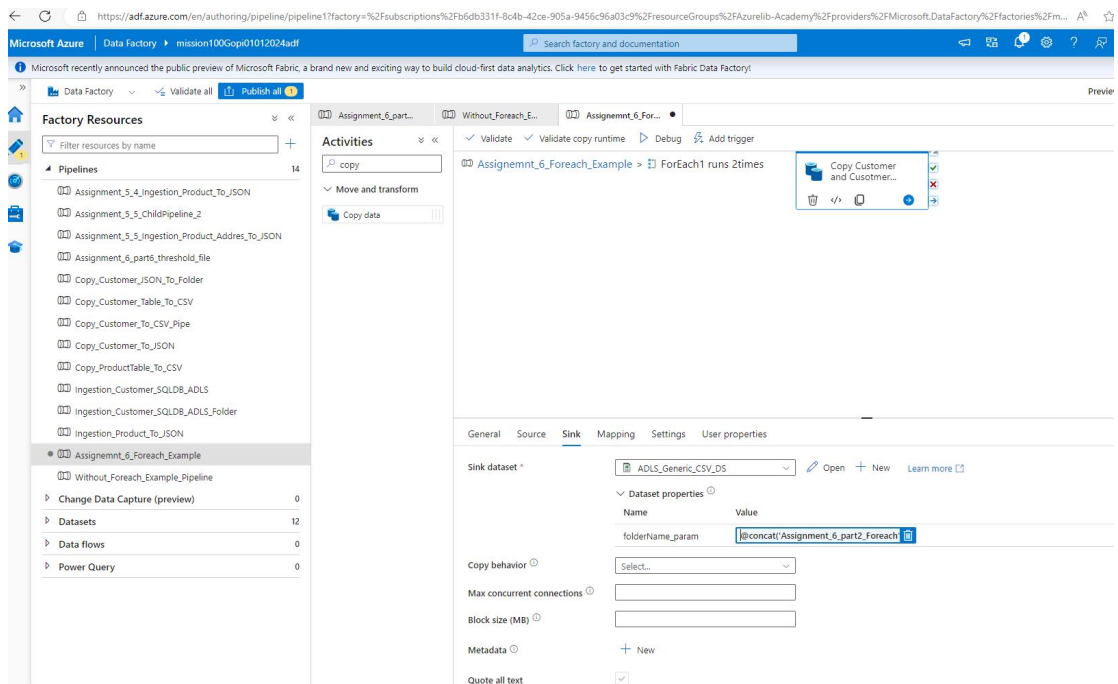
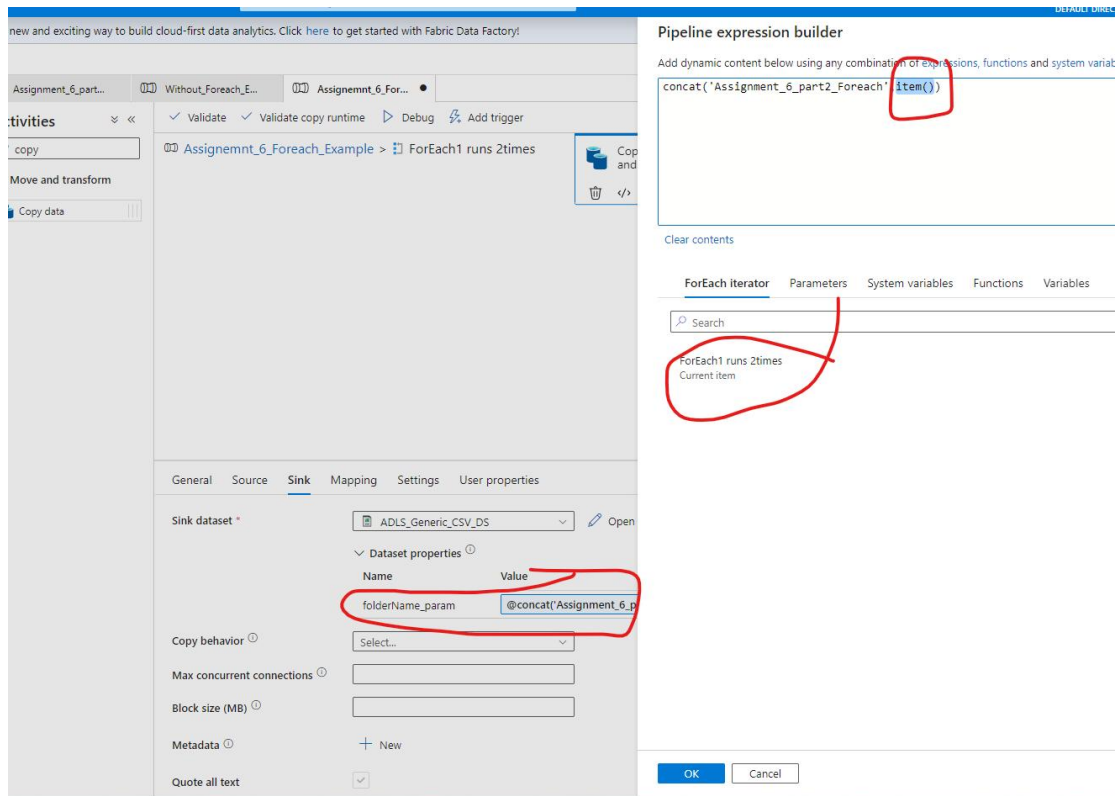
Block size (MB)

Metadata

+ New

Quote all text

☒



Without_Foreach_E...

Assignemnt_6_For...

✓ Validate

✓ Validate copy runtime

▶ Debug

⚡ Add trigger

Assignemnt_6_Foreach_Example > ForEach1 runs 2times

Copy Customer and Cusotmer...

✓

✗

🗑

</>

📄

➡

➡

General

Source

Sink

Mapping

Settings

User properties

Source dataset *

SQL_DB_Generic_DS

🔗 Open

➕ New

🔗 Preview data

Learn more

Dataset properties

Name	Value
schemaName	SalesLT
tableName	@item()

Use query

☒ Table

☐ Query

☐ Stored procedure

Query timeout (minutes)

120

Isolation level

Select...

Partition option

☒ None

☐ Physical partitions of table

☐ Dynamic range

Search factory and documentation

iting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

6_part... Without_Foreach_E... Assignemnt_6_For...

Validate Cancel options Add trigger

transform

ForEach

ForEach1

Activities

Copy Customer...

Parameters Variables Settings **Output**

Pipeline run ID: 707ff802-4e0a-4d14-bbc2-0ef8f500f4d0 **Pipeline status**

All status List

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integratio
Copy Customer and Cusotme...	Queued	Copy data	2/9/2024, 9:28:12 PM	6s	
Copy Customer and Cusotme...	Queued	Copy data	2/9/2024, 9:28:12 PM	6s	
ForEach1	In progress	ForEach	2/9/2024, 9:28:12 PM	7s	

Upload Add Directory Refresh Rename Delete Change tier Acquire lease Break lease Gi

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ Address		
☐ Assignment_5_5_Address_Table_CSV		
☐ Assignment_5_5Copy_Product_Table_JSON		
☐ Assignment_6_part2_ForeachCustomer		
☐ Assignment_6_part2_ForeachCustomerAddress		
☐ Assignment6_part1_Customer_without_Foreach		
☐ Assignment6_part1_CustomerAddress_CSV_without_Foreach		
☐ CSV		
☐ CSV_Pipe		
☐ CSV2		
☐ JSON		

The screenshot shows the Azure Storage Explorer interface. On the left, the 'Upload' tab is active, showing a directory structure with a file named 'SalesLT.CustomerAddress.txt'. The main pane displays the content of this file, which is a CSV-like data set with columns: CustomerID, AddressID, AddressType, rowguid, and ModifiedDate. The data consists of 27 rows of customer information, including IDs, office names (e.g., 'Main Office', 'Shipping'), and timestamps.

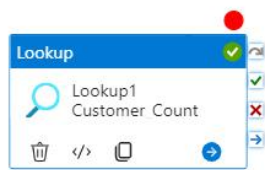
Tricky Questions:

3. Create a pipeline name 'Foreach_Example_2' which solves the following business use case. Customer data is very important for our business. Hence whenever we have more than 100 records in the customer table, we copy the customer data to another table customer_copy within the sql db. However whenever we do this copy, we first truncate the table 'customer_copy' and then copy the data from 'customer' table
Hint : Use ADF to solve this problem.

Solution:

1st Lookup Activity to get customer count, -->
IF (Lookp_Custcount >100)

```
{ Note: In Sink you have to create "Cust_copy"
Pre-script :Truncate Cust_copy table(if exists)
Copy "Cust" to "Cust_copy"
}
```



General **Settings** User properties

Source dataset * SQL_Customer_Table_Dataset Open New Preview data Learn more

First row only ☒

Use query ☐ Table ☒ Query ☐ Stored procedure

Query *

Select COUNT(*) AS CUSTOMER_COUNT
FROM SalesLT.Customer

Edit

Query timeout (minutes) ⓘ

Isolation level ⓘ Select...

Partition option ⓘ ☒ None ☐ Physical partitions of table ⓘ ☐ Dynamic range ⓘ

i Please preview data to validate the partition settings.

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 15
 - Assignment_5_4_Ingestion_Product_To_JSON
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product_Addres_To_JSON
 - Assignment_6_Foreach_Example
 - Assignment_6_part3_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_Folder
 - Ingestion_Product_To_JSON
 - Ass_6_part_3_Foreach_Example_2_Truncate**
 - Without_Foreach_Example_Pipeline
- Change Data Capture (preview) 0
- Datasets 12
- Data flows 0
- Power Query 0

Activities

General

Lookup

Lookup

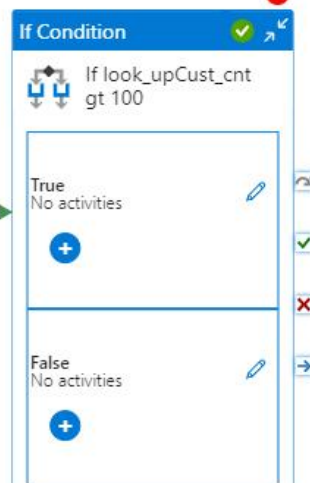
Lookup1 Get Customer count

Output

Copy to clipboard

```
{  "firstRow": {    "CUSTOMER_COUNT": 847  },  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (East US)",  "billingReference": {    "activityType": "PipelineActivity",    "billableDuration": {  }
```

Lookup1 Get Customer count Succeeded Lookup 2/9/202



General Activities (1) User properties

Name * If look_upCust_cnt gt 100 [Learn more](#)

Description we have more than 100 records in the customer table, we copy the customer data to another

Activity state ① ☒ Activated ☐ Deactivated

✓ Validate

▶ Debug

⚡ Add trigger

Lookup

✓

Lookup1
Customer Count

If Condition

✓

⚡

If look_upCust_cnt
gt 100

True

No activities

+

False

No activities

+

General

Activities (1)

User properties

Expression ⓘ

@greater(activity('Lookup1 Customer...

Case	Activity
True	Move customer t... 1 Activity
False	No activities

Pipeline expression builder



Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@greater(activity('Lookup1 Customer_Count').output.firstRow.  
  CUSTOMER_COUNT, 100)
```

[Clear contents](#)

Activity outputs

[Parameters](#)

[System variables](#)

[Functions](#)

[Variables](#)

Lookup1 Customer_Count

Lookup1 Customer_Count activity output

Lookup1 Customer_Count

Lookup1 Customer_Count pipeline return value

Lookup1 Customer_Count first row

Data of the first row

Copy data

 Move customer to customer copy...

General

Source

Sink

Mapping

Settings

User properties

Source dataset *

SQL_DB_Generic_DS

Open

New

Preview data

Learn more

Dataset properties

Name	Value
schemaName	SalesLT
tableName	Customer

Use query

Table

Query

Stored procedure

Query timeout (minutes)

120

Isolation level

Select...

Partition option

None

Physical partitions of table

Dynamic range

«

✓ Validate

✓ Validate copy runtime

▶ Debug

⚡ Add trigger

Ass_6_3_Foreach_Example_2 > If look_upCust_cnt gt 100 > True activities

Copy data

 Move customer to customer_copy...



GeneralSourceSinkMappingSettingsUser properties

schemaNameSalesLt

tableNameCustomer_copy

Write behavior

☒ Insert☐ Upsert☐ Stored procedure

Bulk insert table lock

☐ Yes☒ No

Table option

☐ None☒ Auto create table

Pre-copy script

Specify a script for Copy Activity to execute before writing data into sink table in each run. You can use this property to clean up the pre-loaded data.

e.g. 00:30:00

Write batch size

First Run Pipeline : DO not use TRUNCATE Command

Copy data

 Move customer to customer copy...

General Source Sink Mapping Settings User properties

Dataset properties ⓘ

Name	Value
------	-------

schemaName	SalesLt
------------	---------

tableName	CUSTOMER_COPY
-----------	---------------

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ⓘ ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table ⓘ

Pre-copy script ⓘ

Write batch timeout ⓘ e.g. 00:30:00

rand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

ich_E... •

Validate Validate copy runtime Debug Add trigger

Ass_6_3_Foreach_Example_Z > If look_upCust_cnt gt 100 > True activities

nsform

Copy data

Move customer to customer copy...

General Source Sink Mapping Settings User properties

Dataset properties

Name	Value
schemaName	SalesLt
tableName	CUSTOMER_COPY

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table

Pre-copy script

Write batch timeout e.g. 00:30:00

Second time Run Pipeline with TRUNCATE Command:

<<

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_3_Foreach_Example_2 > ⚙ If look_upCust_cnt gt 100 > True activities

Copy data

 Move customer to customer copyV...



General Source Sink Mapping Settings User properties

Dataset properties ⓘ

Name	Value	Type
schemaName	SalesLt	string
tableName	CUSTOMER_COPY	string

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ⓘ ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table ⓘ

Pre-copy script ⓘ

TRUNCATE TABLE
SALESLT.CUSTOMER_COPY

Add dynamic content [Alt+Shift+D]

« ✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_3_Foreach_Example_2 > If look_upCust_cnt gt 100 > True activities

Copy data

Move customer to customer copy...

General Source Sink Mapping Settings User properties

schemaName SalesLt

tableName CUSTOMER_COPY

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ^① ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table ^①

Add dynamic c

Pre-copy script ^①

TRUNCATE T
SALESLT.CUS

Write batch timeout ^① e.q. 00:30:00

It automatically creates the table (if nonexistent) in source schema, which is not supported when stored procedure is used as write behavior.

**TRUNCATE TABLE
SALESLT.CUSTOMER_COPY**

the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Validate all Publish all

Ass_6_3_Foreach_E...

Activities

cop

Move and transform

Copy data

Lookup

Lookup1 Customer Count

If Condition

If look_upCust_cnt gt 100

True No activities

False No activities

Parameters Variables Settings Output

Pipeline run ID: b8fb53d5-8052-4ad7-a55d-704ac4e44ca9 Pipeline status In progress

All status List

Monitor in Azure Metrics Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User
Lookup1 Customer_Count	In progress	Lookup	2/10/2024, 12:17:40 AM	2s		

Microsoft Azure Data Factory mission100Gopi01012024adf Search factory and documentation

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

Pipelines

- Assignment_5_4_Ingestion_Product_...
- Assignment_5_5_ChildPipeline_2
- Assignment_5_5_Ingestion_Product_...
- Assignment_6_Foreach_Example
- Assignment_6_part6_threshold_file
- Copy_Customer_JSON_To_Folder
- Copy_Customer_Table_To_CSV
- Copy_Customer_To_CSV_Pipe
- Copy_Customer_To_JSON
- Copy_ProductTable_To_CSV
- Ingestion_Customer_SQLDB_ADLS
- Ingestion_Customer_SQLDB_ADLS_F...
- Ingestion_Product_To_JSON
- Ass_6_3_Foreach_Example_2
- Without_Foreach_Example_Pipeline

Change Data Capture (preview) 0

Datasets 12

Data flows 0

Power Query 0

Ass_6_3_Foreach_E...

Activities

cop

Move and transform

Copy data

Ass_6_3_Foreach_Example_2 > If look_upCust_cnt gt 100 > True activities

Copy data

Move customer to customer copy...

General Source Sink Mapping Settings User properties

schemaName SalesLt

tableName CUSTOMER_COPY

Write behavior

Insert Upsert Stored procedure

Bulk insert table lock

Yes No

Table option

None Auto create table

Pre-copy script

TRUNCATE TABLE SALESLT.CUSTOMER_COPY

Write batch timeout

e.g. 00:30:00

Write batch size

https://adf.azure.com/en/authoring/pipeline/pipeline1/rfactory=%2Fsubscriptions%2Fb0b3311-8c4b-42ce-902a-9456c9ba05c9%2Fresourcegroups%2FAzureIIB-Academy%2Fproviders%2FMicrosoft.DataFactory

Microsoft Azure | Data Factory | mission100Gop01012024adf | Search factory and documentation

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

- Pipelines 15
 - Assignment_5_4_Ingestion_Product...
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product...
 - Assignment_6_Foreach_Example
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_F...
 - Ingestion_Product_To_JSON
 - Ass_6_3_Foreach_Example_2
 - Without_Foreach_Example_Pipeline
- Change Data Capture (preview) 0
- Datasets 12
- Data flows 0
- Power Query 0

Activities

Ass_6_3_Foreach_Example_2 > If look_upCust_cnt gt 100 > True activities

Copy data

Copy data

Move customer to customer copy...

General Source Sink Mapping Settings User properties

schemaName SalesLT
tableName CUSTOMER_COPY

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table

Pre-copy script
TRUNCATE TABLE SALESLT.CUSTOMER_COPY

Write batch timeout
e.g. 00:30:00

Write batch size

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

- Pipelines 15
 - Assignment_5_4_Ingestion_Product...
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product...
 - Assignment_6_Foreach_Example
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_F...
 - Ingestion_Product_To_JSON
 - Ass_6_3_Foreach_Example_2
 - Without_Foreach_Example_Pipeline
- Change Data Capture (preview) 0
- Datasets 12
- Data flows 0
- Power Query 0

Activities

Ass_6_3_Foreach_Example_2 > If look_upCust_cnt gt 100 > True activities

Copy data

Copy data

Move customer to customer copy...

General Source Sink Mapping Settings User properties

schemaName SalesLT
tableName CUSTOMER_COPY

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ☐ Yes ☒ No

Table option ☐ None ☒ Auto create table

Pre-copy script
TRUNCATE TABLE SALESLT.CUSTOMER_COPY

Write batch timeout
e.g. 00:30:00

Write batch size

Lookup

Lookup1 Customer Count

If Condition

If look_upCust_cnt gt 100

True No activities

False No activities

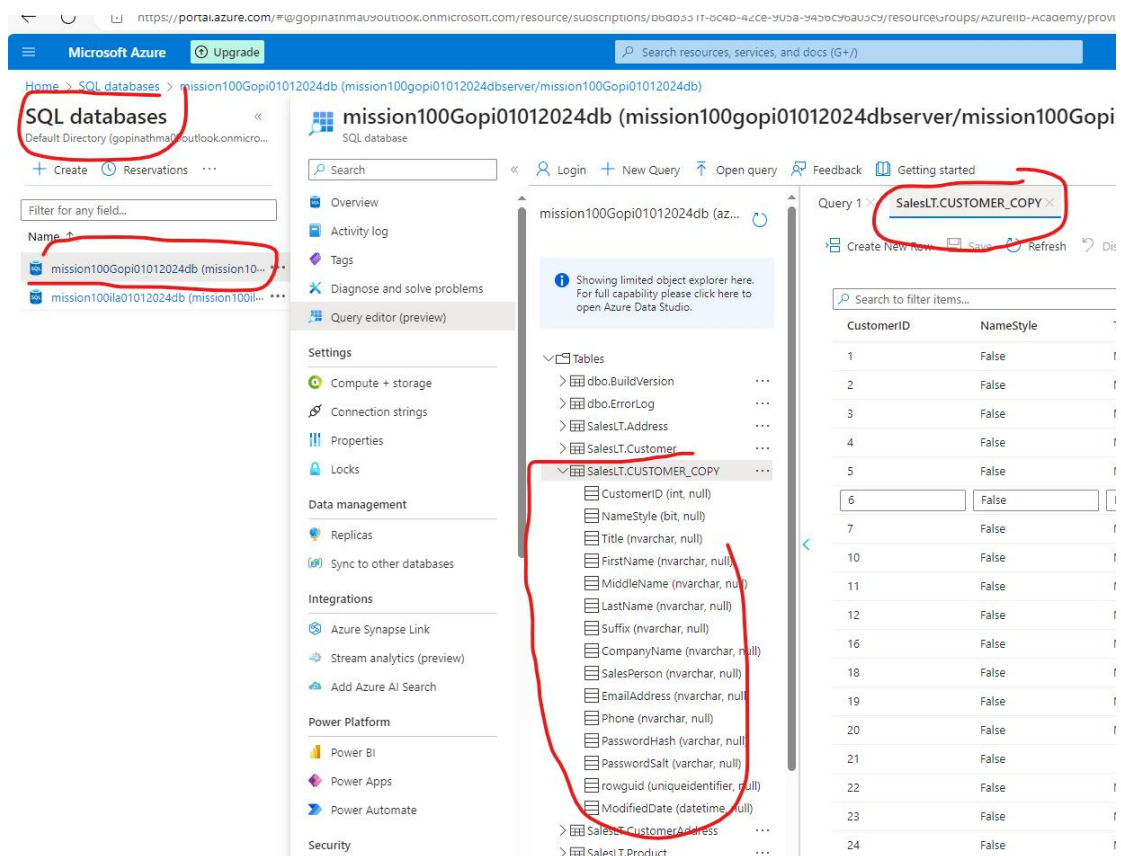
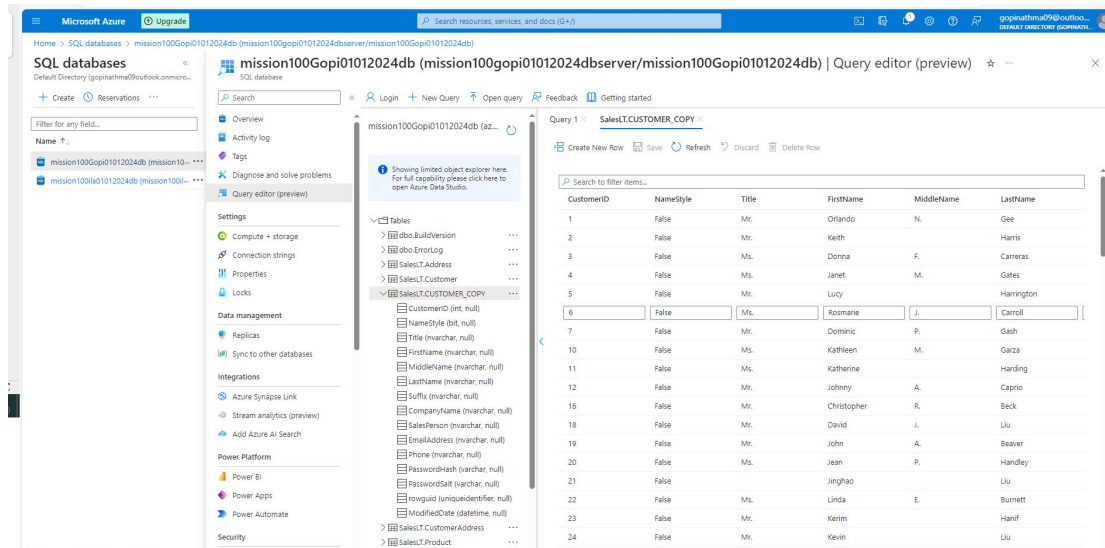
Parameters Variables Settings Output

Pipeline run ID: b8fb53d5-8052-4ad7-a55d-704c4e44ca9 Pipeline status Succeeded View debug run consumption

All status List

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User
Move customer to customer...	Succeeded	Copy data	2/10/2024, 12:17:45 AA	16s	AutoResolveIntegration	
If look_upCust_cnt gt 100	Succeeded	If Condition	2/10/2024, 12:17:45 AA	18s		
Lookup1 Customer_Count	Succeeded	Lookup	2/10/2024, 12:17:40 AA	3s	AutoResolveIntegration	



4. Business wants to move the data from 3 different tables (Customer, Product, CustomerAddress) to ADLS location in CSV format. However they want to make a pipeline in such a manner that All these copy happens through an individual pipeline, which is called by one common pipeline.

Solution: For each with @CreateArray (Customer, Product, CustomerAddress)

Ass_6_4_Foreach_E... • Assignment_6_Fore...

✓ Validate ▶ Debug ⚡ Add trigger

ForEach

ForEach1

Activities

Copy Customer... → +

🗑️ </> 📄 +

General Settings Activities (1) User properties

Sequential ☐

Batch count ⓘ

Items @createArray('Customer','Product','Cust...'

to exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Ass_6_4_Foreach_E... • Assignment_6_Fore...

✓ Validate ▶ Debug ⚡ Add trigger

ForEach

ForEach1

Activities

Copy Customer... → +

🗑️ </> 📄 +

General Settings Activities (1) User properties

Sequential ☐

Batch count ⓘ

Items @createArray('Customer','Product','Cust...'

Pipeline expression builder

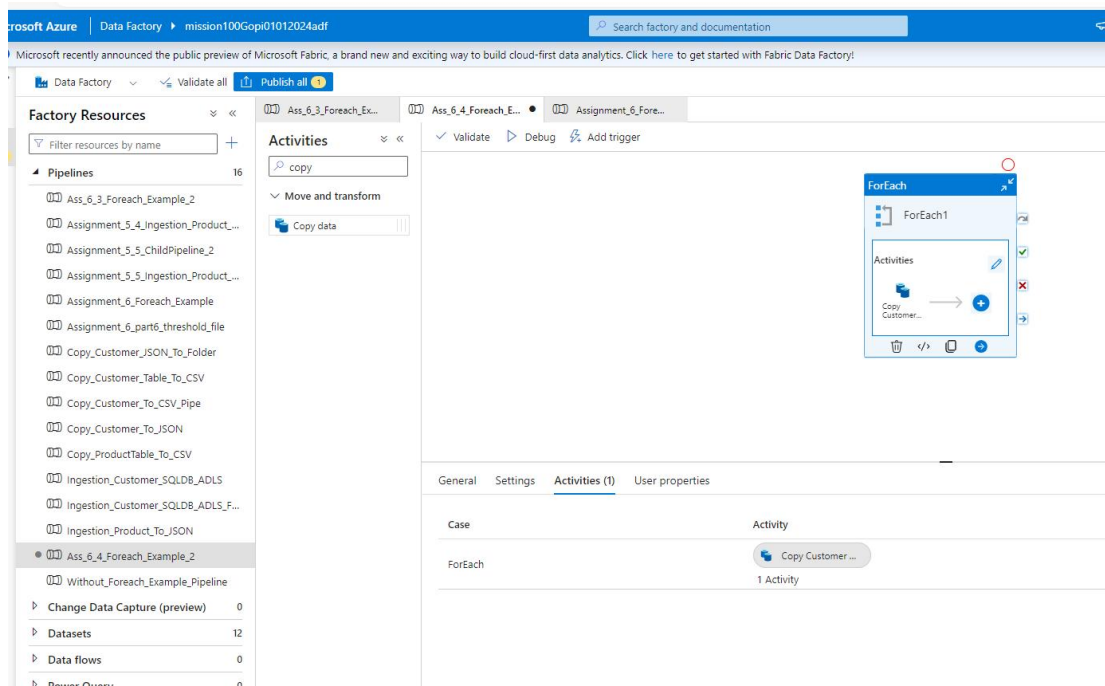
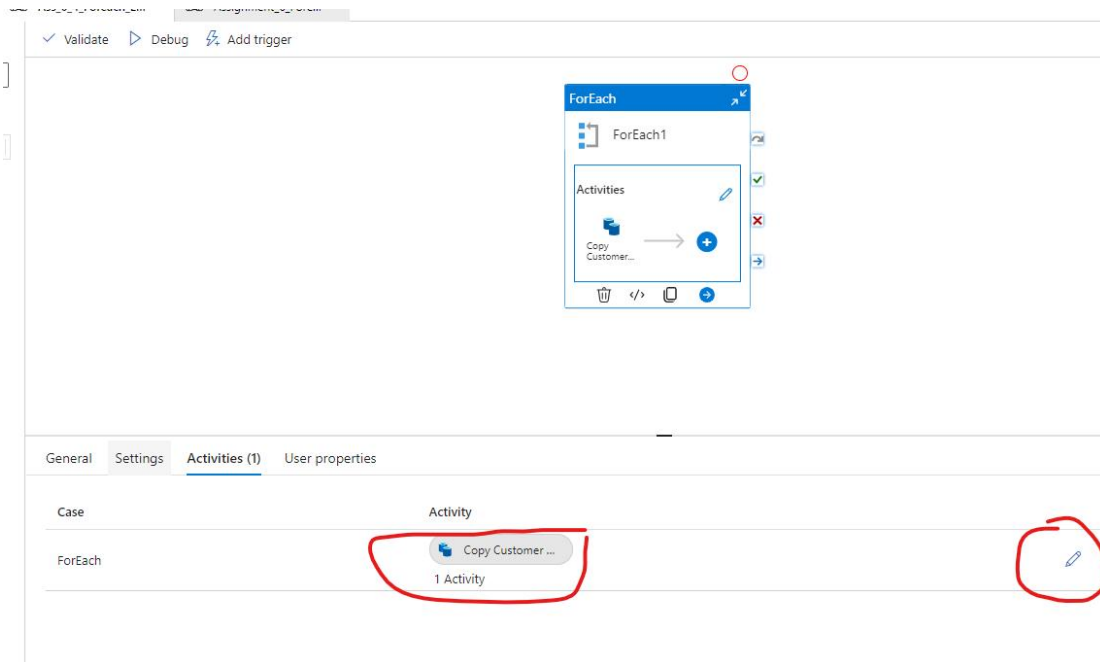
Add dynamic content below using any combination of expressions, functions and system va

```
@createArray('Customer','Product','CustomerAddress')
```

[Clear contents](#)

[Activity outputs](#) [Parameters](#) [System variables](#) [Functions](#) [Variables](#)

Copy Customer Product CustomerAddress
Copy Customer Product CustomerAddress activity output
Copy Customer Product CustomerAddress
Copy Customer Product CustomerAddress pipeline return value



Ass_6_4_Foreach_E...Assignment_6_Fore...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_4_Foreach_Example_2 > ▶ ForEach1

Copy Customer Product...

✓ ✗ ↻ ↵ ➡

GeneralSourceSinkMappingSettingsUser properties

Name *

Copy Customer Product CustomerAddress [Learn more](#)

Description

Business wants to move the data from 3 different tables (Customer, Product, CustomerAddress) to

Activity state ⓘ

☒ Activated ☐ Deactivated

Timeout ⓘ

0.12:00:00

Retry ⓘ

0

Retry interval (sec) ⓘ

30

Ass_6_4_Foreach_E...Assignment_6_Fore...

« ✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_4_Foreach_Example_2 > ▶ ForEach1

Copy Customer Product...

✓ ✗ ↻ ↵ ➡

GeneralSourceSinkMappingSettingsUser properties

Source dataset *

SQL_DB_Generic_DS [Open](#) [New](#) [Preview data](#) [Learn more](#)

Dataset properties ⓘ

Name	Value
schemaName	SalesLT
tableName	@item()

Use query

☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) ⓘ

120

Isolation level ⓘ

Select...

Ass_6_4_Foreach_E... • Assignment_6_Fore...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_4_Foreach_Example_2 > ForEach1

Copy Customer Product...

✖

✓

➔

⌕

</>

🗑

General Source Sink Mapping Settings User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open + New Learn more

Dataset properties

Name	Value
folderName_param	@concat('Ass_6_4_three_Tables_forea...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Ass_6_4_Foreach_E... • Assignment_6_Fore...

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_6_4_Foreach_Example_2 > ForEach1

Copy Customer Product...

✖

✓

➔

⌕

</>

🗑

General Source Sink Mapping Settings User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open + New Learn more

Dataset properties

Name	Value
folderName_param	@concat('Ass_6_4_three_Tables_forea...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Microsoft recently announced the public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click here to get started with Fabric Data Factory!

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 16
 - Ass_6_3_Foreach_Example_2
 - Assignment_5_4_Ingestion_Product...
 - Assignment_5_5_ChildPipeline_2
 - Assignment_5_5_Ingestion_Product...
 - Assignment_6_Foreach_Example
 - Assignment_6_part6_threshold_file
 - Copy_Customer_JSON_To_Folder
 - Copy_Customer_Table_To_CSV
 - Copy_Customer_To_CSV_Pipe
 - Copy_Customer_To_JSON
 - Copy_ProductTable_To_CSV
 - Ingestion_Customer_SQLDB_ADLS
 - Ingestion_Customer_SQLDB_ADLS_F...
 - Ingestion_Product_To_JSON
 - Ass_6_4_Foreach_Example_2
 - Without_Foreach_Example_Pipeline
- Change Data Capture (preview) 0
- Datasets 12
- Data flows 0
- Power Query 0

Activities

copy

Move and transform

Copy data

ForEach

ForEach1

Activities

Copy Customer...

Parameters Variables Settings Output

Pipeline run ID: a4d21adf-6195-40e2-a9f9-Sade3e467a12 Pipeline status In progress

All status List

Showing 1 - 4 of 4 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User property
Copy Customer Product Cust...	Queued	Copy data	2/10/2024, 12:35:34 AM	5s		
Copy Customer Product Cust...	Queued	Copy data	2/10/2024, 12:35:34 AM	5s		
Copy Customer Product Cust...	Queued	Copy data	2/10/2024, 12:35:34 AM	5s		
ForEach1	In progress	ForEach	2/10/2024, 12:35:32 AM	6s		

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Storage accounts > gopio1012024ads | Containers >

landing

Container

Search

Upload Add Directory Refresh Rename Delete Change tier Acquire lease Break lease Give feedback

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing

Search blobs by prefix (case-sensitive) Show deleted objects

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
Address						...
Ass_6_4_three_Tables_foreachCustomer						...
Ass_6_4_three_Tables_foreachCustomerAddress						...
Ass_6_4_three_Tables_foreachProduct						...
Assignment_5_3_Address_Table_CSV						...
Assignment_5_3Copy_Product_Table_JSON						...
Assignment_6_part2_ForeachCustomer						...

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing

Search blobs by prefix (case-sensitive) Show deleted objects

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
Address						-
Ass_6_4_three_Tables_foreachCustomer						-
Ass_6_4_three_Tables_foreachCustomerAddress						-
Ass_6_4_three_Tables_foreachProduct						-
Assignment_5_3_Address_Table_CSV						-

Upload Add Directory Refresh Rename Delete Change tier Acquire lease Break lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / Ass_6_4_three_Tables_foreachCustomerAddress

Search blobs by prefix (case-sensitive)

Name	Modified
[.]	
SalesLt.CustomerAddress.txt	2/10/2024, 12:35:44 AM

[Upload](#) [Add Directory](#) ...

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing /

Ass_6_4_three_Tables_foreachCustomer

Search blobs by prefix (case-...)

☐ Show deleted objects

Name

☐ [.] ...
☒ SalesLt.Customer.txt ...

Ass_6_4_three_Tables_foreachCustomer/SalesLt.Customer.txt ...

Blob

[Save](#) [Discard](#) [Download](#) [Refresh](#) [Delete](#)Overview Versions Edit Generate SAS

```
1 CustomerID,NameStyle,Title,FirstName,MiddleName,LastName,Suffix,CompanyName,SalesPerson,EmailAddress,F
2 1,False,"Mr.,"Orlando","N.,"Gee",,"A Bike Store","adventure-works\pamelao","orlando@adventure-works
3 2,False,"Mr.,"Keith",,"Harris",,"Progressive Sports","adventure-works\david8","keith@adventure-works
4 3,False,"Ms.,"Donna",,"F.,"Carreras",,"Advanced Bike Components","adventure-works\jillian0","donna@e
5 4,False,"Ms.,"Janet",,"M.,"Gates",,"Modular Cycle Systems","adventure-works\jillian0","janet1@adventu
6 5,False,"Mr.,"Lucy",,"Harrington",,"Metropolitan Sports Supply","adventure-works\shu0","lucy@adventu
7 6,False,"Ms.,"Rosmarie",,"J.,"Carroll",,"Aerobic Exercise Company","adventure-works\linda3","rosmarie
8 7,False,"Mr.,"Dominic",,"P.,"Gash",,"Associated Bikes","adventure-works\shu0","dominic@adventure-wor
9 10,False,"Ms.,"Kathleen",,"M.,"Garza",,"Rural Cycle Emporium","adventure-works\josel1","kathleen@adve
10 11,False,"Ms.,"Katherine",,"Harding",,"Sharp Bikes","adventure-works\josel1","katherine@adventure-wor
11 12,False,"Mr.,"Johnny",,"A.,"Caprio",,"Jr.,"Bikes and Motorbikes","adventure-works\garrett1","johnny@
12 16,False,"Mr.,"Christopher",,"R.,"Beck",,"Jr.,"Bulk Discount Store","adventure-works\jaeo","christoph
13 18,False,"Mr.,"David",,"J.,"Liu",,"Catalog Store","adventure-works\michael9","david2@adventure-works
14 19,False,"Mr.,"John",,"A.,"Beaver",,"Center Cycle Shop","adventure-works\pamelao","john@adventure-wc
15 20,False,"Ms.,"Jean",,"P.,"Handley",,"Central Discount Store","adventure-works\david8","jean@adventu
16 21,False,"Jinghao",,"Liu",,"Chic Department Stores","adventure-works\jillian0","jinghao1@adventure-wc
17 22,False,"Ms.,"Linda",,"E.,"Burnett",,"Travel Systems","adventure-works\jillian0","linda4@adventure-w
18 23,False,"Mr.,"Kerim",,"Hanif",,"Bike World","adventure-works\shu0","kerim@adventure-works.com","21f
19 24,False,"Mr.,"Kevin",,"Liu",,"Eastside Department Store","adventure-works\linda3","kevin5@adventure-
```

[Upload](#) [Add Directory](#) ...

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing /

Ass_6_4_three_Tables_foreachCustomerAddress

Search blobs by prefix (case-...)

☐ Show deleted objects

Name

☐ [.] ...
☒ SalesLt.CustomerAddress.txt ...

Ass_6_4_three_Tables_foreachCustomerAddress/SalesLt.CustomerAddress.txt ...

Blob

[Save](#) [Discard](#) [Download](#) [Refresh](#) [Delete](#)Overview Versions Edit Generate SAS

```
1 CustomerID,AddressID,AddressType,rowguid,ModifiedDate
2 29485,1086,"Main Office",16765338-dbe4-4421-b5e9-3836b9278e63,2007-09-01 00:00:00.0000000
3 29486,621,"Main Office",22b3e910-14af-4ed5-8b4d-23bb757414d,2005-09-01 00:00:00.0000000
4 29489,1069,"Main Office",a095c88b-d7e6-4178-a078-2eca44214801,2005-07-01 00:00:00.0000000
5 29490,887,"Main Office",f12e1702-d897-4035-b614-0fe2c72168a9,2006-09-01 00:00:00.0000000
6 29492,618,"Main Office",5b3b3eb2-3f43-47ed-a20c-23697dabf23b,2006-12-01 00:00:00.0000000
7 29494,537,"Main Office",492d92b6-31af-47ea-8e00-7c11c7ccc20d,2005-09-01 00:00:00.0000000
8 29496,1072,"Main Office",0a66b0f3-24bc-4148-9661-26e935cec99a,2007-09-01 00:00:00.0000000
9 29497,889,"Main Office",7e0b56fd-7324-4898-bedb-2f56a843bb0f,2005-07-01 00:00:00.0000000
10 29499,527,"Main Office",c90cb0c3-976a-4075-a2b9-13136f4f1a92,2006-09-01 00:00:00.0000000
11 29502,893,"Main Office",8acb1c6a-7cdf-417b-afd5-9f4c642f3c7e,2007-07-01 00:00:00.0000000
12 29503,32,"Shipping",be00ffc6-194b-44e9-a129-dfb0ebf1301f,2006-08-01 00:00:00.0000000
13 29503,541,"Main Office",6811dbe3-1f02-440c-932e-a54f403b3e51,2006-08-01 00:00:00.0000000
14 29505,1083,"Main Office",740a06af-77f1-45ca-b284-38537324f86a,2007-05-01 00:00:00.0000000
15 29506,1082,"Main Office",dd51a14d-45f5-44e9-bf3b-260f0e08e545,2007-08-01 00:00:00.0000000
16 29508,619,"Main Office",2dd326f6-02da-44b7-96fb-b3a86067c5ee,2005-12-01 00:00:00.0000000
17 29510,540,"Main Office",af6dcc27-dccb-40de-9445-3f4990aabd28,2005-08-01 00:00:00.0000000
```

[Upload](#) [Add Directory](#) ...

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing /

Ass_6_4_three_Tables_foreachProduct

Search blobs by prefix (case-...)

☐ Show deleted objects

Name

☐ [.] ...
☒ SalesLt.Product.txt ...

Ass_6_4_three_Tables_foreachProduct/SalesLt.Product.txt ...

Blob

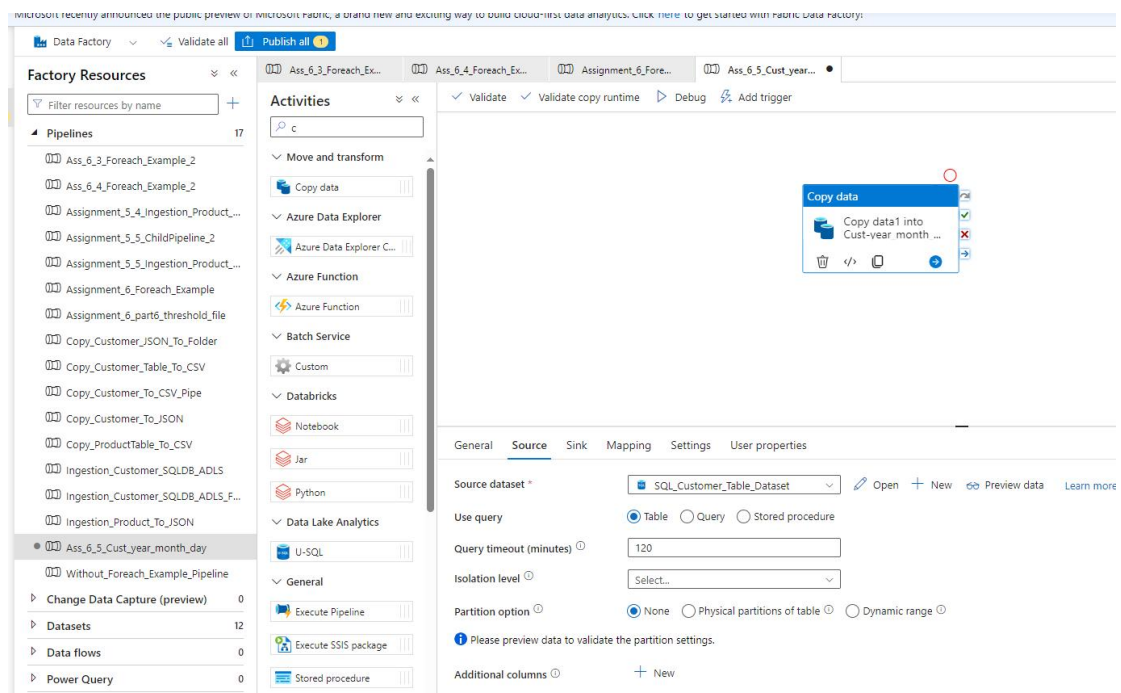
[Save](#) [Discard](#) [Download](#) [Refresh](#) [Delete](#)Overview Versions Edit Generate SAS

```
1 ProductID,Name,ProductNumber,Color,StandardCost,ListPrice,Size,Weight,ProductCategoryID,ProductModelID
2 600,"HL Road Frame - Black, 58","FR-R92R-58","Black",1059.3100,1431.5000,"58",1016.04,18,6,2002-06-01 00:00:00.0000000
3 706,"HL Road Frame - Red, 58","FR-R92R-58","Red",1059.3100,1431.5000,"58",1016.04,18,6,2002-06-01 00:00:00.0000000
4 707,"Sport-100 Helmet, Red","HL-U509-R","Red",13.0863,34.9900,,35,33,2005-07-01 00:00:00.0000000,,0x
5 708,"Sport-100 Helmet, Black","HL-U509","Black",13.0863,34.9900,,35,33,2005-07-01 00:00:00.0000000,,0x
6 709,"Mountain Bike Socks, M","SO-B909-M","White",3.3963,9.5000,"M",27,18,2005-07-01 00:00:00.0000000,,0x
7 710,"Mountain Bike Socks, L","SO-B909-L","White",3.3963,9.5000,"L",27,18,2005-07-01 00:00:00.0000000,,0x
8 711,"Sport-100 Helmet, Blue","HL-U509-B","Blue",13.0863,34.9900,,35,33,2005-07-01 00:00:00.0000000,,0x
9 712,"AWC Logo Cap","CA-1098","Multi",6.9223,8.9900,,23,2,2005-07-01 00:00:00.0000000,,0x474946383961
10 713,"Long-Sleeve Logo Jersey, S","LJ-0192-S","Multi",38.4923,49.9900,"S",25,11,2005-07-01 00:00:00.0000000,,0x
11 714,"Long-Sleeve Logo Jersey, M","LJ-0192-M","Multi",38.4923,49.9900,"M",25,11,2005-07-01 00:00:00.0000000,,0x
12 715,"Long-Sleeve Logo Jersey, L","LJ-0192-L","Multi",38.4923,49.9900,"L",25,11,2005-07-01 00:00:00.0000000,,0x
13 716,"Long-Sleeve Logo Jersey, XL","LJ-0192-X","Multi",38.4923,49.9900,"XL",25,11,2005-07-01 00:00:00.0000000,,0x
14 717,"HL Road Frame - Red, 62","FR-R92R-62","Red",868.6342,1431.5000,"62",1043.26,18,6,2005-07-01 00:00:00.0000000
15 718,"HL Road Frame - Red, 44","FR-R92R-44","Red",868.6342,1431.5000,"44",961.61,18,6,2005-07-01 00:00:00.0000000
16 719,"HL Road Frame - Red, 48","FR-R92R-48","Red",868.6342,1431.5000,"48",979.75,18,6,2005-07-01 00:00:00.0000000
17 720,"HL Road Frame - Red, 52","FR-R92R-52","Red",868.6342,1431.5000,"52",997.90,18,6,2005-07-01 00:00:00.0000000
18 721,"HL Road Frame - Red, 56","FR-R92R-56","Red",868.6342,1431.5000,"56",1016.04,18,6,2005-07-01 00:00:00.0000000
19 722,"LL Road Frame - Black, 58","FR-R38B-58","Black",204.6251,337.2200,"58",1115.83,18,9,2005-07-01 00:00:00.0000000
20 723,"LL Road Frame - Black, 60","FR-R38B-60","Black",204.6251,337.2200,"60",1124.90,18,9,2005-07-01 00:00:00.0000000
21 724,"LL Road Frame - Black, 62","FR-R38B-62","Black",204.6251,337.2200,"62",1133.98,18,9,2005-07-01 00:00:00.0000000
```

Solution:

Customer/2024/01/28/10/03/00/salesltcustomer.text

Customer/{formatDateTime(utcnow(),'yyyy')}



Ass_6_4_Foreach_Ex...Assignment_6_Fore...Ass_6_5_Cust_year...

ValidateValidate copy runtimeDebugAdd trigger

Copy data

Copy data1 into Cust-year_month ...

✓✗→

GeneralSinkMappingSettingsUser properties

Sink dataset *

ADLS_Generic_CSV_DS

OpenNewLearn more

Dataset properties

Name	Value
folderName_param	value Add dynamic content [Alt+Shift+D]

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

+ New

Search factory and documentation

gopinathma09@ou

build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

te

✓ Validate copy runtime

▶ Debug

⚙ Add trigger

Copy data

Copy cust data to
yyyymmdd

Source

Sink

Mapping

Settings

User properties

set *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name

Value

folderName_param

Customer/@{formatDateTime(utcno...

avior

Select...

urrent connections

MB

New

Quote all text

✓

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system

Customer/@{formatDateTime(utcnow(),'yyyy')}/@{formatDateTime(utcnow(),'MM')}/@{formatDateTime(utcnow(),'dd')}

Clear contents

Parameters

System variables

Functions

Variables

Search

OK

Cancel

Search factory and documentation

gopinathma09@ou

build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

✓ Validate

✓ Validate copy runtime

▶ Debug

⚙ Add trigger

Copy data

Copy cust data to
yyyymmdd

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name

Value

folderName_param

Customer/@{formatDateTime(utcno...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

New

Quote all text

✓

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

Customer/@{formatDateTime(utcnow(),'yyyy')}/@{formatDateTime(utcnow(),'MM')}/@{formatDateTime(utcnow(),'dd')}

Clear contents

Parameters

System variables

Functions

Variables

format

Collapse all

Date Functions

formatDateTime

Returns a string in date format. The result is a string in ISO 8601 format ("o") by default, unless a fo...

adddays

Adds an integer number of days to a string timestamp passed in. The number of days can be posi...

addhours

Adds an integer number of hours to a string timestamp passed in. The number of hours can be po...

addminutes

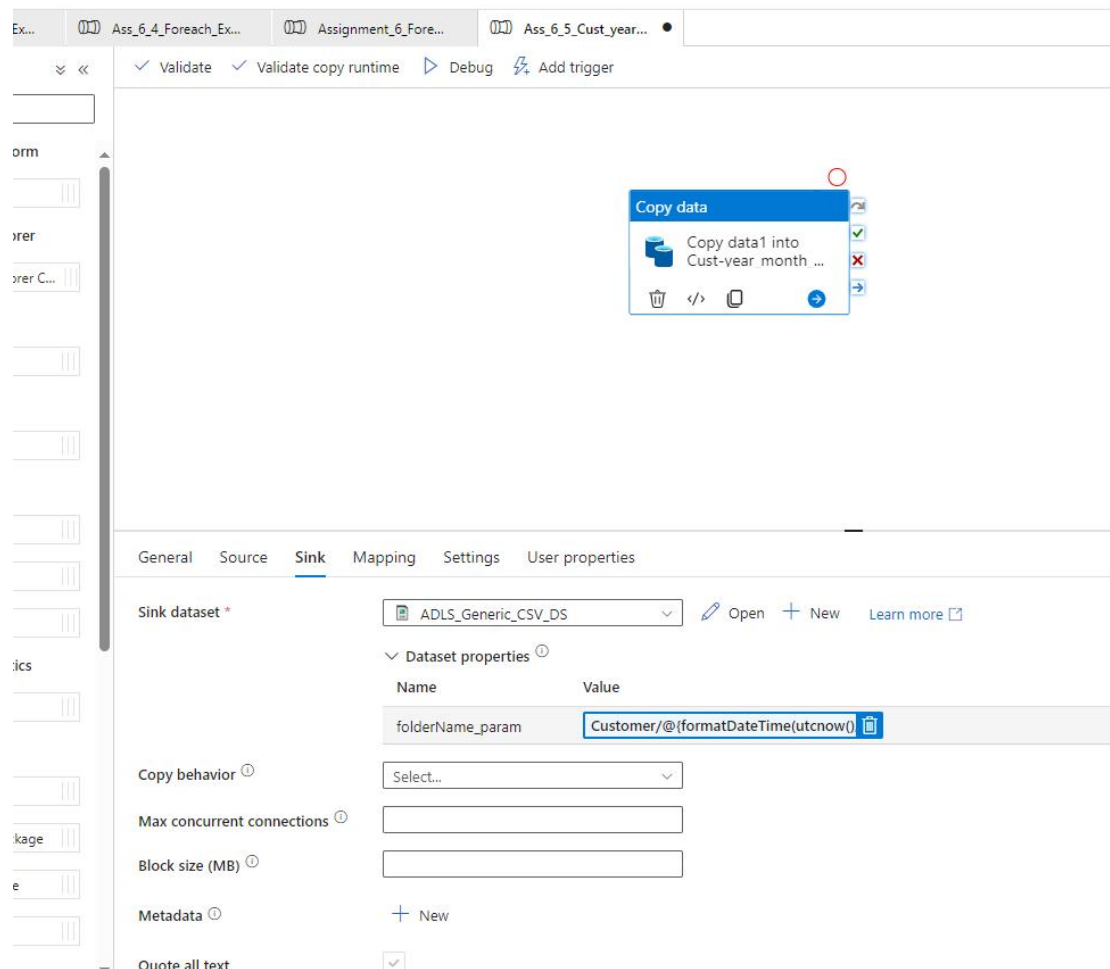
Adds an integer number of minutes to a string timestamp passed in. The number of minutes can b...

addseconds

Adds an integer number of seconds to a string timestamp passed in. The number of seconds can b...

OK

Cancel



Customer/@{formatDateTime(utcnow(), 'yyyy')}/@{formatDateTime(utcnow(), 'MM')}/@{formatDateTime(utcnow(), 'dd')}

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
Customer/@{formatDateTime(utcnow(), 'yyyy')}/@{formatDateTime(
  utcnow(), 'MM')}/@{formatDateTime(utcnow(), 'dd')}/@
  {formatDateTime(utcnow(), 'hh')}/@{formatDateTime(utcnow(),
    'mm')}/@{formatDateTime(utcnow(), 'ss')}
```

[Clear contents](#)

[Parameters](#)

[System variables](#)

[Functions](#)

[Variables](#)

portal.azure.com/?quickstart=true#view/Microsoft_Azure_Storage/ContainerMenuBlade/~/overview/storageAccountId/%2Fsubscriptions%2Fb6db331f-8c4b-42ce-905a-9456c96a03c9%2FresourceGroups...

Microsoft Azure UpgradeSearch resources, services, and docs (G+)

home > Storage accounts > gop101012024adls | Containers >

landingContainer

Search

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak leaseGive feedback

OverviewDiagnose and solve problemsAccess Control (IAM)

SettingsShared access tokensManage ACLAccess policyPropertiesMetadata

Authentication method: Access key (Switch to Microsoft Entra user account)Location: landing / Customer / 2024 / 02 / 10Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	
[-]				B
SalesLT.Customer.txt	2/10/2024, 1:36:29 AM	Hot (Inferred)		B

Ass_6_4_Foreach_Ex...Assignment_6_Fore...Ass_6_5_Cust_year...

ValidateCancel optionsAdd trigger

Copy dataCopy data1 into Cust-year month ...

ParametersVariablesSettingsOutput

Pipeline run ID: 23e8892e-c251-4bec-808d-8017dc0f0666Pipeline

All status

Showing 1 - 1 of 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integ
Copy data1 into Cust-year_m...	Queued	Copy data	2/10/2024, 12:59:17 AM	7s	

landing

Container

- Search
- Overview
- Diagnose and solve problems
- Access Control (IAM)

- Settings
- Shared access tokens
- Manage ACL
- Access policy
- Properties
- Metadata

Upload Add Directory Refresh Rename Delete Change tier Acquire lease Break lease Give feedback

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / Customer / 2024 / 02 / 10

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	
☐ [-]				B
☐ SalesLT.Customer.txt	2/10/2024, 1:36:29 AM	Hot (Inferred)		BI